

Dissecting Causal Mechanism Shifts via FANS: Function And Noise Separation

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Abstract

Identifying the drivers of causal mechanism shifts, distinguishing functional changes from noise alterations, termed *dissection*, is a critical yet under-explored problem in data science (e.g., biomedical science and manufacturing). This paper introduces a more general and unified framework, the *function and noise separation* (FANS) framework, that detects and dissects shifts in non-additive, non-linear Structural Causal Models (SCMs) beyond existing additive noise models. Our approach is grounded in a theoretical independence criterion, where function shifts induce a statistical dependence between a node’s parents and residual noise. Building on this foundation, we develop a practical two-stage algorithm to efficiently detect and dissect these shifts without retraining. Furthermore, we address the complex challenge of simultaneous function and noise shifts, introducing a formal assumption to resolve their inherent non-identifiability. Our results are corroborated by simulations. Our code is available at <https://github.com/MLAI-Yonsei/FANS/>.

1. Introduction

Observed data frequently originates from diverse environments (Mooij et al., 2020; Karlsson & Krijthe, 2023), leading to conditional distribution changes known as causal mechanism shifts (CMS) (Chen et al., 2023). A causal mechanism is the conditional probability of a node given its parent nodes (Mameche et al., 2024; Perry et al., 2022) in a directed acyclic graph (DAG) describing the data generating processes. A CMS is an alteration in this conditional probability. This phenomenon is examined in multiple disciplines, such as *structural breaks* in econometrics (Bai, 1996; Perron

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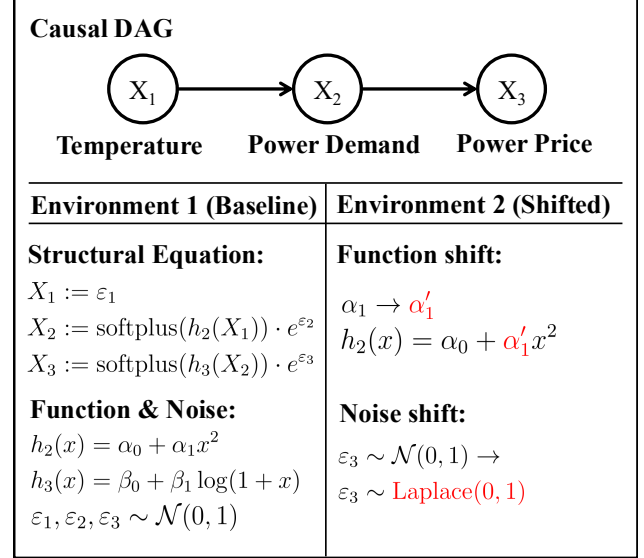


Figure 1. Causal graphs depicting how causal relationships vary under different shifts.

et al., 2020) and *conditional distribution tests* in statistics (Hu & Lei, 2024).

Detecting a CMS amounts to pinpointing the node exhibiting a distributional shift in its mechanism across environments. While the detection of CMS and its applications (Reddy & N Balasubramanian, 2024; Mameche et al., 2024; Perry et al., 2022) have been well-addressed (Chen et al., 2023; Mameche et al., 2022), their drivers have received less focus. A CMS can arise from two sources: a shift in the deterministic *causal function* relating variables, or a shift in the distribution of the *exogenous noise* component. A function shift implies that the fundamental relationship between variables has changed. In contrast, a noise shift suggests a change in the influence of unobserved latent variables (Pearl, 2009), warranting investigation into how the underlying context has varied.

Figure 1 illustrates a simple three-node structural equation model for an electricity market with temperature (X_1), power demand (X_2), and power price (X_3). In Environment 1, all nodes follow the same nonlinear, non-additive form. Environment 2 differs from Environment 1 in two ways. First, the causal function from temperature to de-

mand changes: the coefficient α_1 in h_2 is replaced by α'_1 , so that the temperature–demand relationship itself (the increase in demand per degree of temperature) is modified. We call this a function shift. Second, the noise distribution of the price node changes from Gaussian to Laplace, while the function remains fixed. This represents a noise shift: the demand–price curve is unchanged, but new, demand-independent shocks—for example, forecast errors from newly introduced renewable energy sources—make price residuals more heavy-tailed.

Distinguishing these sources is crucial, as misinterpretation can lead to flawed conclusions. Consider the simple electricity market model in Figure 1. Suppose that the causal function (h) remains unchanged across two countries or time periods, but a new energy source (e.g., large-scale wind and solar generation) is introduced. Its unpredictable forecast errors generate rare but large price spikes, so that the noise distribution changes from approximately Gaussian to a heavier-tailed law such as Laplace. If one ignores this noise shift and keeps assuming Gaussian errors, standard estimation procedures may contort h to absorb these spikes and falsely suggest that the demand–price sensitivity itself has changed. In other words, a change in volatility driven by new energy sources may be incorrectly interpreted as a change in the causal function, leading to misleading policy or market conclusions.

In this paper, we propose a novel framework for dissecting causal mechanism shifts. Specifically, our main contributions are threefold.

First, we relax the restrictive additive noise assumption to accommodate a broader class of causal mechanism. Our method employs Normalizing Flows (NF) to infer a noise component analogous to a residual, applicable to non-additive, non-linear models.

Second, we propose a two-stage framework for detecting and dissecting shifts. The framework trains once on the base environment; new environments require only inference and an independence test.

Third, we address the non-identifiability of simultaneous shifts, where both function and noise change concurrently. We establish formal structural assumptions that make it theoretically possible to distinguish and dissect drivers.

1.1. Related Work

Our work addresses the challenge of comparing observational data from two distinct environments, assuming the data are generated by SCMs that differ in their functional relationships or noise distribution. This problem intersects with several lines of research.

Dissecting Causal Mechanism Shifts. Dissecting a CMS

involves attributing a change in a conditional distribution to a function shift or a noise shift. This task is simplified under the additive noise model assumption: changes in the conditional mean are attributed to the function, while changes in variance or higher-order moments are attributed to the noise. LinearCCP (Huang et al., 2024) assumes a linear, additive structure, enabling the use of beta-coefficient tests for function shifts and two-sample tests on residuals for noise shifts. Extending to non-linear cases, iSCAN (Chen et al., 2023) can identify functionally shifted edges but still relies on the ANM assumption and an additional inter-parent additivity constraint. The kernel-based approach in (Li et al., 2023) tests for functional changes within the ANM framework but may impose further constraints like Gaussian noise and can be computationally intensive in high-dimensional settings.

Detecting Causal Mechanism Shifts. A broader line of research focuses on detecting the presence of a CMS without dissecting its cause. At the individual node level, this is treated as a two-sample testing problem for conditional distributions (Lee et al., 2024; Yan et al., 2022), addressed by various non-parametric tests and deep generative models (Kostic et al., 2024; Chang et al., 2024). However, many approaches, including recent conditional two-sample tests (Hu & Lei, 2024; Chen & Lei, 2025; Chatterjee et al., 2024), require per-node examination of the conditional distribution within the DAG, leading to high computational complexity.

Another relevant area is Intervention Target Estimation (ITE), which aims to efficiently identify which nodes in a causal graph have been affected by interventions. Early approaches in the linear and soft-intervention setting, such as CITE (Varici et al., 2021), exploit changes in the precision matrix to scalably recover targets without learning the full DAG. Follow-up work, such as PreDITEr (Varici et al., 2022), extends this framework to handle latent confounding, demonstrating that one can still characterize and estimate valid intervention targets even without full structural recovery. More recently, methods such as LIT (Yang et al., 2024) broaden the scope to heterogeneous and nonlinear environments, proposing a two-step recovery and matching procedure to pinpoint unknown targets. Deep learning-based methods, such as DeepITE (Tao et al., 2024), leverage graph autoencoders and auxiliary labels to jointly learn across multiple intervention instances, thereby improving scalability and generalization. Overall, while these methods have made significant progress in detecting intervention targets, they do not explicitly dissect the nature of these changes. In contrast, the central goal of our framework is to perform both detection and dissection simultaneously. Furthermore, our approach can analyze local mechanism shifts at specific values of parent variables, enabling a more focused and targeted investigation of where and how a shift has occurred.

2. Preliminaries

We are using the structural causal models (Pearl, 2009) as a framework for data generating mechanisms across environments: a base environment (1) and a new environment (2), where a causal mechanism shift has occurred. Given the data from both environments and the causal DAG from the base environment, our goal is to identify the nodes affected by the shift and classify their type as either a function shift or a noise shift.

Definition 2.1 (SCM (Pearl, 2009)). A structural causal model (SCM) $\mathcal{M} = (f, P_\varepsilon)$ is composed of a collection of J structural equations:

$$X_j := f_j(\text{PA}_j, \varepsilon_j), \quad j \in \{1, \dots, J\}. \quad (1)$$

Here, P_ε is a joint distribution over the noise variables, where ε_i and ε_j for $i, j \in \{1, \dots, J\}$ are jointly independent. ε_j is an exogenous variable and PA_j denotes parent variables of node X_j , independent of ε_j . The structural equations are determined by a vector of functions $f = [f_1, \dots, f_J]$. Since we assume the independence condition for noise variables, the set of observed variables is causally sufficient.

To leverage the framework of Causal Normalizing Flows (CNF) (Javaloy et al., 2023), we make the following assumptions.

Assumption 2.2. For each $j \in \{1, \dots, J\}$, the function f_j is a diffeomorphism; i.e., for any fixed realization of $\text{PA}_j = \text{pa}_j$, the function f_j mapping from ε_j to X_j is bijective, and both the mapping and its inverse are differentiable.

Definition 2.3 (Environment). Each environment $k \in \{1, 2\}$ is characterized by a data distribution $P_X^{(k)}$ generated by an SCM $\mathcal{M}^{(k)}$. We denote the random variable X_j as observed in environment k by $X_j^{(k)}$. Different environments $k \neq k'$ are distinguished by the fact that at least one causal mechanism in $\mathcal{M}^{(k)}$ differs from the corresponding one in $\mathcal{M}^{(k')}$.

Furthermore, for the application of our method, we introduce an assumption on the structure of the causal mechanisms:

Assumption 2.4. Each structural equation f_j can be decomposed as:

$$f_j(\text{PA}_j, \varepsilon_j) = g_j(h_j(\text{PA}_j), \varepsilon_j), \quad (2)$$

where $h_j : \mathbb{R}^{d_j} \rightarrow \mathbb{R}^{k_j}$ is a function of the parents only, and $g_j : \mathbb{R}^{k_j} \times \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable link function that combines their outputs. g_j is a link function invariant across environments. Variations in PA_j are absorbed solely through h_j , which means that changes in PA_j are restricted to those that can be fully explained by modifications in h_j .

The SCM in each environment $\mathcal{M}^{(k)}$ is defined as follows:

$$X_j^{(k)} := f_j^{(k)}(\text{PA}_j^{(k)}, \varepsilon_j^{(k)}) = g_j(h_j^{(k)}(\text{PA}_j^{(k)}), \varepsilon_j^{(k)}). \quad (3)$$

Definition 2.5 (Causal Mechanism Shift). (Chen et al., 2023; Mameche et al., 2024) A causal mechanism shift at j refers to a situation where the conditional distributions of a variable X_j given its parents differ between two environments:

$$P(X_j^{(1)} | \text{PA}_j^{(1)}) \neq P(X_j^{(2)} | \text{PA}_j^{(2)}). \quad (4)$$

For example, Figure 1 shows X_2 and X_3 experience causal mechanism shifts. Let the causal graph for environment 1 be $\mathcal{G}^{(1)}$. We assume that the causal structure in environment 2 results from interventions on $\mathcal{G}^{(1)}$. It implies that the parent set for each variable X_j in environment 2 is a subset of the parent set for X_j in environment 1.

Assumption 2.6. For a node j that doesn't experience a shift, the conditional probabilities remain invariant.

$$P(X_j^{(1)} | \text{PA}_j^{(1)}) = P(X_j^{(2)} | \text{PA}_j^{(2)}).$$

Definition 2.7 (Shifted Nodes). Indices j for variables X_j that experience a causal mechanism shift are referred to as shifted nodes, denoted by the set S :

$$S = \{1 \leq j \leq J : P(X_j^{(1)} | \text{PA}_j^{(1)}) \neq P(X_j^{(2)} | \text{PA}_j^{(2)})\}.$$

A shift across environments is characterized by $S \neq \emptyset$. The shift can be categorized into two types:

Assumption 2.8 (Shift Types). Environment 2 differs from environment 1 in one of the following ways for a node X_j : a function shift ($h_j^{(1)} \neq h_j^{(2)}$) or a noise shift ($\varepsilon_j^{(1)} \stackrel{d}{\neq} \varepsilon_j^{(2)}$).

We use the notation $\stackrel{d}{=}$ and $\stackrel{d}{\neq}$ to denote equality and inequality in distribution, respectively. The function difference is defined as $\Delta h_j(\mathbf{pa}) := h_j^{(2)}(\mathbf{pa}) - h_j^{(1)}(\mathbf{pa})$ for any given input value $\mathbf{pa}$. A function shift covers cases where an incoming edge to $X_j^{(2)}$ is removed. This corresponds to the new function $h_j^{(2)}$ mapping the input from that parent to a constant value. Since a simultaneous shift of function and noise is often indistinguishable from a shift in the function alone, we analyze shifts as two separate cases for identifiability. Section 3.3 provides an example illustrating the non-identifiability and presents a strategy to make the shifts identifiable.

Function shifts and noise shifts are not mutually exclusive; i.e., some shifts may reside in the intersection between function and noise shifts. For example, consider the following observed shifts:

$$X_j^{(1)} = \text{PA}_j^{(1)} \varepsilon_j^{(1)}, \quad X_j^{(2)} = 2\text{PA}_j^{(2)} \varepsilon_j^{(2)}.$$

This shift can be viewed as a functional shift by transforming PA_j to $2PA_j$; or a noise shift which transforms ε_j to $2\varepsilon_j$. For such shifts that can be considered as function and noise shifts, throughout the paper, we will consider it as a noise-shift. Formally, let $\mathcal{T}_{g_j}$ be the set of functional variations $\delta_j : \mathbb{R}^{d_j} \rightarrow \mathbb{R}^{k_j}$ such that the composition with the link function can be factored into a noise transformation:

$$\mathcal{T}_{g_j} := \{\delta_j \mid \exists \psi : \mathbb{R} \rightarrow \mathbb{R} \text{ such that } \psi \text{ is invertible and } \forall \mathbf{pa}, g_j(h_j^{(1)}(\mathbf{pa}) + \delta_j(\mathbf{pa}), \varepsilon_j^{(1)}) \stackrel{d}{=} g_j(h_j^{(1)}(\mathbf{pa}), \psi(\varepsilon_j^{(1)}))\}.$$

If a shift is categorized as a function shift ($h_j^{(2)} = h_j^{(1)} + \Delta h_j$), then the difference term satisfies $\Delta h_j \notin \mathcal{T}_{g_j}$. If $\Delta h_j \in \mathcal{T}_{g_j}$, we categorize it as a noise shift due to non-identifiability. As a caveat, we make sure that this shift is different from simultaneous shift such as

$$X_j^{(1)} = PA_j^{(1)} \varepsilon_j^{(1)}, \quad X_j^{(2)} = \exp(PA_j^{(2)})(2\varepsilon_j^{(1)}).$$

For any SCM that induces a DAG, we can express observed data by exogenous variables. We denote $X^{(k)} = [X_1^{(k)}, \dots, X_J^{(k)}]$ and $\varepsilon^{(k)} = [\varepsilon_1^{(k)}, \dots, \varepsilon_J^{(k)}]$. For environment 1, $X^{(1)}$ is expressed as follows:

$$\begin{aligned} [X_1^{(1)}, \dots, X_J^{(1)}] &= [f_1^{(1)}(PA_1^{(1)}, \varepsilon_1^{(1)}), \dots, f_J^{(1)}(PA_J^{(1)}, \varepsilon_J^{(1)})] \\ &= [\tilde{f}_1^{(1)}(\varepsilon_{\text{an}_1}^{(1)}), \dots, \tilde{f}_J^{(1)}(\varepsilon_{\text{an}_J}^{(1)})] \\ &= \tilde{f}^{(1)}([\varepsilon_1^{(1)}, \dots, \varepsilon_J^{(1)}]) = \tilde{f}^{(1)}(\varepsilon^{(1)}). \end{aligned}$$

Since all parent variables consist of noise variables, $\varepsilon_{\text{an}_j}^{(1)}$ denotes the noise ancestors that determine $X_j^{(1)}$. Equivalently, by unrolling the recursive definitions (Javaloy et al., 2023) of each $f_j^{(1)}$, we obtain a non-recursive map $\tilde{f}^{(1)} : \mathbb{R}^J \rightarrow \mathbb{R}^J$ such that

$$X^{(1)} = \tilde{f}^{(1)}(\varepsilon^{(1)}). \quad (5)$$

Definition 2.9 (Causal Normalizing Flow). (Javaloy et al., 2023) Causal Normalizing Flow (CNF) $T(X)$ is an autoregressive normalizing flow (ANF) (Kingma et al., 2016; Papamakarios et al., 2017) to learn an invertible transformation from i.i.d random variables $X = [X_1, \dots, X_J]$ to a set of independent base-noise variables $N = [N_1, \dots, N_J]$.

$$T(X) = T(\tilde{f}(\varepsilon)) = [N_1, \dots, N_J]. \quad (6)$$

Base-noise variables are random variables that follow simple base distributions, such as the standard normal distribution. According to Theorem 1 in (Javaloy et al., 2023), for each $X_j \in X$, there exists an invertible function q_j between true noise ε_j from a structural causal model and base noise N_j in a causal normalizing flow, where both generate the same observational data:

$$q_j(\varepsilon_j) = N_j, \quad \forall j \in \{1, \dots, J\}. \quad (7)$$

In our framework, CNF is trained on the observed dataset $X^{(1)}$ from environment 1, denoted as $\hat{T}^{(1)}$. We denote $\hat{T}^{(1)}(X^{(1)})_j$ as the j -th element of $\hat{T}^{(1)}(X^{(1)})$. The trained CNF yields the estimated noise $\hat{N}^{(1)} = \hat{T}^{(1)}(X^{(1)})$. $\hat{T}^{(1)}$ is then applied to the data from environment 2; i.e., $\hat{T}^{(1)}(X^{(2)})$.

3. Method

We assume that Assumptions 2.2-2.8 hold in this section. Our method leverages a single Causal Normalizing Flow (CNF) model, trained only on data from the base environment (environment 1), to perform a comprehensive analysis of causal mechanism shifts.

First, in Section 3.1, the trained CNF is used to estimate the conditional probability distributions for each node under both environments. By measuring the divergence between these distributions, we detect the set of shifted nodes, S . For the nodes in S , Section 3.2 details the dissection process. We input the data from environment 2 into the same CNF model to generate a residual noise, which captures the discrepancy between the original mechanism and the new data. A statistical test for independence between this residual noise and the parents in environment 2 then allows us to classify the cause: independence indicates a noise shift, while dependence implies a function shift. Finally, Section 3.3 addresses the more subtle challenge of non-identifiability between a function-only shift and a simultaneous function-and-noise shift, and we present a set of formal assumptions under which they become distinguishable.

While the existence of a true mapping $q : \varepsilon^{(1)} \rightarrow N^{(1)}$ is theoretically guaranteed by Theorem 1 in (Javaloy et al., 2023), in practice, it is estimated from finite samples. We denote the approximation of this mapping, implemented as a CNF $\hat{T}^{(1)}$, by $\hat{q}$. We assume that $\hat{q}$ is invertible.

3.1. Detection of Shift

To detect a shift for a node X_j between environments 1 and 2, we test the null hypothesis that their conditional probability distributions are identical:

$$H_{0,j} : P(X_j^{(1)} \mid PA_j^{(1)}) = P(X_j^{(2)} \mid PA_j^{(2)})$$

To test this hypothesis, we leverage the model $\hat{T}^{(1)}$. The core idea is to assess how well $\hat{T}^{(1)}$ explains $X^{(2)}$ by simulating a counterfactual distribution on data from environment 1. This is based on the do-operator in CNF (Javaloy et al., 2023).

Let $P_{\text{do}(PA_j=c)}^{(1)}$ be the counterfactual distribution generated by applying the $\text{do}(PA_j = c)$ to Environment 1 data using the model $\hat{T}^{(1)}$. Let $\tilde{P}_{\text{do}(PA_j=c)}^{(2)}$ be the counterfactual

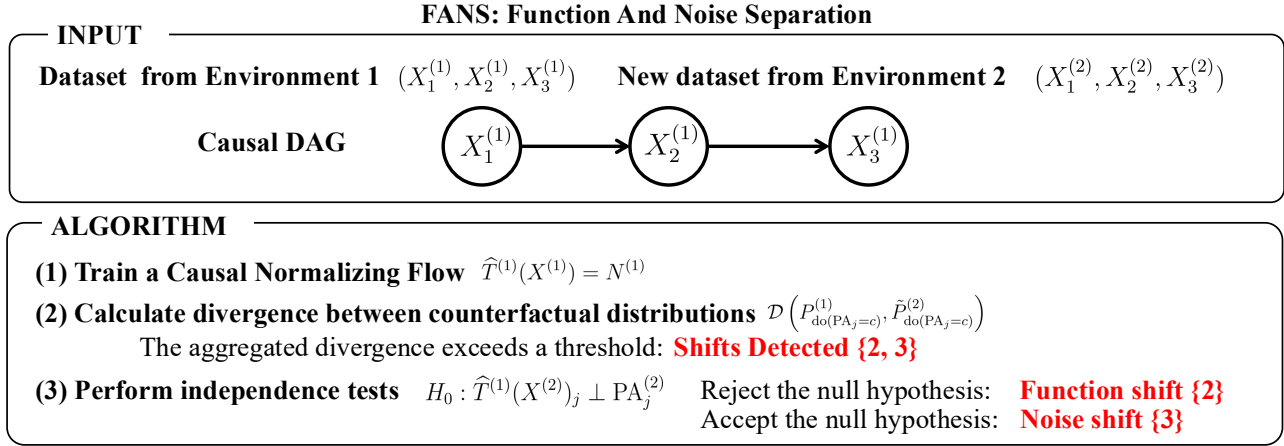


Figure 2. Assuming the data follows the Structural Equation Model (SEM) described in Figure 1, the set of true shifted nodes is $\{2, 3\}$, where X_2 undergoes a function shift and X_3 undergoes a noise shift. This figure illustrates the required inputs and the algorithmic procedure of FANS to detect and dissect these shifts.

distribution generated by applying the same intervention procedure to Environment 2 data, also using the model $\hat{T}^{(1)}$.

Let c be a point in the support of the parent variables $\text{PA}_j^{(1)}$. The local divergence at point c is defined as the divergence between the two local conditional distributions:

$$\mathcal{D}_{j,c} := \mathcal{D}(P_{\text{do}(\text{PA}_j=c)}^{(1)}, \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)}) \quad (8)$$

where $\mathcal{D}$ is a divergence measure. $\mathcal{D}$ is any divergence satisfying $\mathcal{D}(P, Q) \geq 0$ and $\mathcal{D}(P, Q) = 0$ iff $P = Q$. Let the test statistic, $\bar{\mathcal{D}}_j$, be the empirical average of this local divergence over n points $\{c_1, \dots, c_n\}$ sampled from the parent distribution $P(\text{PA}_j)$. As the number of samples n approaches infinity, $\bar{\mathcal{D}}_j$ converges to the expected divergence:

$$\mathbb{E}_{c \sim P(\text{PA}_j^{(1)})}[\mathcal{D}_{j,c}]$$

Theorem 3.1 (Shift Detection Equivalence). *A causal mechanism shift at node X_j has occurred if and only if the expected divergence is strictly greater than zero.*

$$\begin{aligned} \mathbb{E}_{c \sim P(\text{PA}_j^{(1)})}[\mathcal{D}_{j,c}] > 0 &\iff \\ P(X_j^{(1)} | \text{PA}_j^{(1)}) &\neq P(X_j^{(2)} | \text{PA}_j^{(2)}) \end{aligned}$$

A node is identified as shifted if this aggregated value exceeds a predefined threshold τ :

$$S = \{j : \bar{\mathcal{D}}_j > \tau\}. \quad (9)$$

For the index $j \in S$, identified as shifted, we perform a dissection to determine whether the shift is primarily due to a function shift or a noise shift.

3.2. Dissection of Shift

To understand the type of shift, we start with a question: what noise would explain the new observation $X_j^{(2)}$ if we

apply the base model from environment 1? We refer to this hypothetical noise as the residual noise $\tilde{\epsilon}_j^{(2)}$. It acts as a proxy that captures the discrepancy between the original mechanism and the new data.

$$X_j^{(2)} = g_j(h_j^{(1)}(\text{PA}_j^{(2)}), \tilde{\epsilon}_j^{(2)}).$$

Due to our assumption that $f_j^{(1)}$ is invertible, the solution is uniquely determined by:

$$\tilde{\epsilon}_j^{(2)} = [(f_j^{(1)})^{-1}(X_j^{(2)})]_j.$$

We infer noise from $X^{(2)}$ by $\hat{T}^{(1)}$. The learned transformation $\hat{T}^{(1)}$ maps the data to noise:

$$\hat{T}^{(1)}(X^{(1)}) = \hat{q}(\epsilon^{(1)}) = \hat{q}((f^{(1)})^{-1}(X^{(1)})).$$

By applying this transformation to the new data $X^{(2)}$, we obtain the transformed residual noise:

$$\hat{T}^{(1)}(X^{(2)}) = \hat{q}((f^{(1)})^{-1}(X^{(2)})) = \hat{q}(\tilde{\epsilon}^{(2)}). \quad (10)$$

Theorem 3.2 (Independence Criterion). *For a shifted node X_j , the statistical independence of its residual noise $\tilde{\epsilon}_j^{(2)}$ and its parents $\text{PA}_j^{(2)}$ indicates the type of shift:*

$$\begin{aligned} \tilde{\epsilon}_j^{(2)} \not\perp \text{PA}_j^{(2)} &\iff \text{the shift is a function shift.} \\ \tilde{\epsilon}_j^{(2)} \perp \text{PA}_j^{(2)} &\iff \text{the shift is a noise shift.} \end{aligned}$$

A fundamental property of statistical independence is that it is preserved under transformations like $\hat{q}$. The conclusion of Theorem 3.2 holds for both the original and transformed noise.

$$\tilde{\epsilon}_j^{(2)} \perp \text{PA}_j^{(2)} \iff \hat{q}_j(\tilde{\epsilon}_j^{(2)}) \perp \text{PA}_j^{(2)}.$$

Algorithm 1 FANS: Function And Noise Separation for Dissecting Causal Mechanism Shifts

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1: Input:  $X^{(1)}, X^{(2)}$ , and  $\mathcal{G}^{(1)}$ .
2: Output:  $\hat{S}$ , and for each  $j \in \hat{S}$ , the identified shift type (Function, Noise).
3: Phase 1: Learning the Base Causal Mechanism
4: Train CNF  $\hat{T}^{(1)}$  on  $X^{(1)}$ .
5: Phase 2: Shift Detection
6: Initialize  $\hat{S} \leftarrow \emptyset$ .
7: for each node index  $j$  do
8:   Evaluate the divergence  $\mathcal{D}_{j,c}$  (Eq. 8)
9:   Aggregate over  $c$  to compute  $\bar{\mathcal{D}}_j$ 
10:  if  $\bar{\mathcal{D}}_j > \tau$  then
11:    add  $j$  to the set of shifted nodes:  $\hat{S} \leftarrow \hat{S} \cup \{j\}$ .
12:  end if
13: end for
14: Phase 3: Shift Dissection
15: for each index for shifted nodes  $j \in \hat{S}$  do
16:    $\hat{q}_j(\tilde{\varepsilon}_j^{(2)}) \leftarrow \hat{T}^{(1)}(X^{(2)})_j$  (Eq. 10)
17:   Perform a statistical independence test for:
      
$$H_{0,j}^{\text{indep}} : \hat{q}_j(\tilde{\varepsilon}_j^{(2)}) \perp \text{PA}_j^{(2)}$$

18:   if  $H_{0,j}^{\text{indep}}$  is rejected at level  $\alpha$  then
19:     Label the shift in  $j$  as a Function Shift.
20:   else
21:     Label the shift in  $j$  as a Noise Shift.
22:   end if
23: end for
24: return  $\hat{S}$  and the labels for each node in  $\hat{S}$ .
    
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This allows us to dissect the shift type by analyzing the independence properties of the transformed noise obtained from the CNF.

To test $H_{0,j}^{\text{indep}}$, we use distance correlation as the test statistic to quantify the degree of dependence. We calculate the distance correlation between $\hat{q}_j(\tilde{\varepsilon}_j^{(2)})$ and $\text{PA}_j^{(2)}$, and reject the test hypothesis if this distance correlation exceeds a certain threshold. Permutation tests based on distance covariances are also applicable.

3.3. Non-Identifiability between Function-Only and Simultaneous Shifts

A key challenge arises when a shift involves the function, as a function-only shift can be observationally indistinguishable from a simultaneous shift of both function and noise. The following example illustrates the non-identifiability.

We consider a structural equation $X_j = g_j(h_j(\text{PA}_j), \varepsilon_j)$ with an additive link function $g_j(h_j(\text{PA}_j), \varepsilon) = a_1 + a_2 \cdot \varepsilon$, where $h_j(\text{PA}_j) = [a_1, a_2]^\top$. Let $\sigma(\cdot)$ denote the sigmoid

function to ensure strict invertibility ($a_2 > 0$). Our base environment is as follows:

$$X_j^{(1)} = \text{PA}_j^{(1)} + \sigma(\text{PA}_j^{(1)}) \cdot \varepsilon_j^{(1)}, \quad \varepsilon_j^{(1)} \sim U(0, 1).$$

Here, $U(\cdot, \cdot)$ denotes a uniform distribution. Data comes from two environments:

$$\text{(Case 1)} \quad h_j^{(2)}(\text{PA}_j) = [\sigma(\text{PA}_j^{(2)}), \sigma(\text{PA}_j^{(2)})]^\top$$

$$X_j^{(2)} = \sigma(\text{PA}_j^{(2)}) + \sigma(\text{PA}_j^{(2)})\varepsilon_j^{(2)}, \quad \varepsilon_j^{(2)} \sim U(0, 1)$$

$$\text{(Case 2)} \quad h_j^{(2)}(\text{PA}_j) = [1.5\sigma(\text{PA}_j^{(2)}), \sigma(\text{PA}_j^{(2)})]^\top$$

$$X_j^{(2)} = 1.5\sigma(\text{PA}_j^{(2)}) + \sigma(\text{PA}_j^{(2)})\varepsilon_j^{(2)}, \quad \varepsilon_j^{(2)} \sim U(-\frac{1}{2}, \frac{1}{2})$$

Both cases yield the same conditional distribution $X_j^{(2)} | \text{PA}_j^{(2)} \sim U(\sigma(\text{PA}_j^{(2)}), 2\sigma(\text{PA}_j^{(2)}))$. Since the scale term is strictly positive, the invertibility assumption (Assumption 2.2) holds in both scenarios. However, the data alone cannot distinguish whether the shift originated solely from the function parameters (Case 1) or from a combination of function scaling and noise reduction (Case 2). We find that if Assumption 3.3 is satisfied, function shifts and function-noise shifts become identifiable.

Assumption 3.3 (Generalized Affine Interaction). Let $g_j^*(h_j^{(1)}, N_j) := g_j(h_j^{(1)}, q_j^{-1}(N_j)) = g_j(h_j^{(1)}, \varepsilon_j)$. We assume there exist real-valued functions $\beta_{j,0}(h)$ and $\beta_{j,1}(h)$ such that the following relationship holds for the function shift:

$$g_j^*(h_j, N_j) = \Psi_j(\beta_{j,1}(h_j) \cdot N_j + \beta_{j,0}(h_j)).$$

$\Psi_j : \mathbb{R} \mapsto \mathbb{R}$ is an invertible and differentiable scalar function that wraps the affine combination. This assumption posits that the function g_j^* exhibits a specific structure. Under the assumptions, we can isolate the effect of the function shift Δh in $q_j(\tilde{\varepsilon}_j^{(2)})$. After standardizing the noise by removing conditional moments, the resulting distribution is shown to be independent of the parents $\text{PA}_j^{(2)}$. This leads to the following theorem.

Theorem 3.4 (Shift Identifiability). *Let $\tilde{q}_j(\tilde{\varepsilon}_j^{(2)})$ be the standardized inferred noise:*

$$\tilde{q}_j(\tilde{\varepsilon}_j^{(2)}) = \frac{q_j(\tilde{\varepsilon}_j^{(2)}) - \mathbb{E}[q_j(\tilde{\varepsilon}_j^{(2)}) | \text{PA}_j^{(2)}]}{\sqrt{\text{Var}(q_j(\tilde{\varepsilon}_j^{(2)}) | \text{PA}_j^{(2)})}}.$$

Under Assumption 3.3 and assuming the CNF base noise follows a standard normal distribution, if the causal mechanism shift is function-only, then the squared standardized noise follows a Chi-squared distribution with one degree of freedom.

$$(\tilde{q}_j(\tilde{\varepsilon}_j^{(2)}))^2 \sim \chi^2(1).$$

The converse holds when a function-noise shift causes $q_j(\varepsilon_j^{(2)})$ to deviate from a normal distribution. The full derivation is provided in the appendix, and Algorithm 1 presents the pseudo-code of FANS.

3.4. Finite-Sample Consistency of the Dissection Test

In finite-sample settings, the estimated CNF map $\hat{q}$ introduces approximation errors, resulting in a non-zero baseline dependence. To address a non-zero baseline bias, our statistic $\Delta\hat{\mathcal{R}}$ computes the absolute difference in distance correlations between the two environments. We show that, under standard moment conditions, the empirical dissection statistic is consistent: function and noise shifts are correctly classified with probability tending to one as the sample size grows.

Let $Z^{(k)} := \text{PA}_j^{(k)}$ denote the parents in environment k , and let $\mathcal{R}(\cdot, \cdot)$ denote the distance correlation (Székely et al., 2007). Define

$$\Delta\mathcal{R}^* := |\mathcal{R}(N^{(2)}, Z^{(2)}) - \mathcal{R}(N^{(1)}, Z^{(1)})|, \quad (11)$$

$$\Delta\hat{\mathcal{R}} := |\mathcal{R}(\hat{N}^{(2)}, Z^{(2)}) - \mathcal{R}(\hat{N}^{(1)}, Z^{(1)})|. \quad (12)$$

By Theorem 3.2, $\Delta\mathcal{R}^* = 0$ under a pure noise shift and $\Delta\mathcal{R}^* > 0$ under a function shift.

Assumption 3.5 (L^2 Estimation Consistency). For each j and $k \in \{1, 2\}$, $\mathbb{E}[(\hat{N}_j^{(k)} - N_j^{(k)})^2] \leq \delta_n^2$, with $\delta_n \rightarrow 0$ as $n \rightarrow \infty$.

Assumption 3.6 (Finite Second Moments). $\mathbb{E}[(N_j^{(k)})^2] < \infty$ and $\mathbb{E}[\|Z^{(k)}\|^2] < \infty$.

Proposition 3.7 (Consistency of the Dissection Statistic). Under Assumptions 3.5 and 3.6,

$$\Delta\hat{\mathcal{R}} \xrightarrow{p} \Delta\mathcal{R}^* \quad \text{as } n \rightarrow \infty. \quad (13)$$

Proof sketch. Cauchy–Schwarz bounds the distance covariance error as $|\mathcal{V}^2(\hat{N}, Z) - \mathcal{V}^2(N, Z)| \leq 8C_Z\delta_n$, where $C_Z := \sqrt{\mathbb{E}[\|Z - Z'\|^2]}$. The continuous mapping theorem, applied to the distance correlation functional, then yields convergence in probability of $\Delta\hat{\mathcal{R}}$ to $\Delta\mathcal{R}^*$. Full details are provided in Appendix B.5.

Corollary 3.8 (Consistent Classification). Fix any threshold $\tau_{dis} \in (0, \Delta\mathcal{R}^*)$ when $\Delta\mathcal{R}^* > 0$. Under Assumptions 3.5 and 3.6:

- If the shift is a pure noise shift ($\Delta\mathcal{R}^* = 0$), then $\Pr(\Delta\hat{\mathcal{R}} > \tau_{dis}) \rightarrow 0$ as $n \rightarrow \infty$.
- If the shift is a function shift ($\Delta\mathcal{R}^* > 0$), then $\Pr(\Delta\hat{\mathcal{R}} > \tau_{dis}) \rightarrow 1$ as $n \rightarrow \infty$.

Proposition 3.7 complements the population-level identifiability of Theorem 3.2 with a finite-sample guarantee: as the

CNF training set grows, the probability of misclassifying a noise shift as a function shift, or vice versa, vanishes. This provides theoretical grounding for the empirical robustness observed across sample sizes and model capacities in our experiments.

4. Experiments

4.1. Synthetic data

We evaluate the model’s ability to detect causal mechanism shifts in Table 1. Baseline methods for comparison include iSCAN (Chen et al., 2023), LinearCCP (Huang et al., 2024), LCIT (Duong & Nguyen, 2022), PreDITer (Varici et al., 2022) and SplitKCI (Pogodin et al., 2024). The data generation process follows (Chen et al., 2023). Initially, 30 random DAGs were generated for each of two graph types: Erdős–Rényi (ER) and Scale-Free (SF). We also randomly selected $0.2J$ shifted nodes. The number of nodes J within each DAG was set to 10, 20, 30, 40, and 50.

We generated 50,000 data points for each case. Some baseline methods were evaluated on a 1,000 sample subset due to computational constraints. For the detection task, iSCAN and SplitKCI were limited by memory complexity. LCIT was limited by time complexity. For the dissection task, GPR was restricted by time complexity. iSCAN was also limited to 1,000 samples due to excessive Type I error. For FANS, all 50,000 samples are used for training and validation, while both the detection and dissection analyses are performed on the same 1,000 sample subset used by the baselines. A Causal Mechanism Shift (CMS) was introduced between environments 1 and 2. The SCM for environment 1 is defined as:

$$X_j^{(1)} := \sum_k \sin([\text{PA}_j^{(1)}]_k^2) + \sigma\left(\sum_k ([\text{PA}_j^{(1)}]_k)\right) \cdot \varepsilon_j^{(1)}.$$

Here, $\sigma(\cdot)$ denotes a sigmoid function and $[\text{PA}_j^{(1)}]_k$ denotes k -th element of the parents $\text{PA}_j^{(1)}$ with $\varepsilon_j \sim \mathcal{N}(0, 1)$. A function shift can manifest in two distinct ways. The first is a change from a $\sin(\cdot)$ function to a $\cos(2(\cdot)^2 - 3(\cdot))$ function. The second is a shift where the set of causal parents is altered by the deletion of a parent. Separately, a noise shift occurs when the distribution of the exogenous noise term is modified. This can take the form of a variance change, such as $\varepsilon_j \sim \mathcal{N}(0, 2)$, or a change in the distribution’s family to $\varepsilon_j \sim \text{Laplace}(0, 1)$.

Each node j in the graph can only exhibit one of three cases: no shift, a function shift, or a noise shift. Table 1 evaluates the model’s ability to identify which nodes have undergone any type of shift. Table 1 indicates that the FANS framework is effective for the task of shifted node detection, outperforming established baseline methods in the tested configurations. Its robust performance, particularly on SF

Table 1. F1-scores for **Shifted Nodes Detection**, detailed by graph type and node count. Results are shown across varying node counts (J) for Erdős-Rényi (ER) and Scale-Free (SF) graphs. Higher F1-scores (F1 $\uparrow$) indicate better performance. Our proposed method consistently outperforms all baseline methods across all experimental settings. Numbers in parentheses indicate standard deviations.

Method	ER Graph - Nodes (J)					SF Graph - Nodes (J)				
	10	20	30	40	50	10	20	30	40	50
iSCAN	0.371 (0.327)	0.484 (0.245)	0.552 (0.149)	0.527 (0.077)	0.553 (0.117)	0.321 (0.330)	0.546 (0.233)	0.357 (0.197)	0.433 (0.170)	0.478 (0.157)
SplitKCI	0.428 (0.248)	0.406 (0.173)	0.431 (0.141)	0.402 (0.118)	0.417 (0.100)	0.295 (0.227)	0.301 (0.108)	0.293 (0.097)	0.320 (0.086)	0.301 (0.067)
LCIT	0.468 (0.199)	0.468 (0.149)	0.500 (0.085)	0.537 (0.063)	0.519 (0.069)	0.353 (0.039)	0.338 (0.026)	0.342 (0.015)	0.328 (0.020)	0.328 (0.023)
LinearCCP	0.731 (0.153)	0.784 (0.117)	0.753 (0.102)	0.765 (0.089)	0.772 (0.077)	0.719 (0.172)	0.714 (0.149)	0.661 (0.134)	0.623 (0.106)	0.609 (0.121)
PreDITEr	0.940 (0.115)	0.978 (0.050)	0.954 (0.077)	0.965 (0.057)	0.963 (0.040)	0.933 (0.136)	0.976 (0.054)	0.959 (0.073)	0.947 (0.067)	0.972 (0.049)
FANS (Ours)	0.989 (0.061)	1.000 (0.000)	0.980 (0.043)	0.982 (0.030)	0.975 (0.037)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)

Table 2. F1-scores for **Shift Type Dissection** across ER/SF graphs and node counts (J).

Method	ER Graph - Nodes (J)					SF Graph - Nodes (J)				
	10	20	30	40	50	10	20	30	40	50
iSCAN	0.611 (0.351)	0.687 (0.250)	0.678 (0.224)	0.645 (0.236)	0.683 (0.198)	0.722 (0.304)	0.620 (0.297)	0.664 (0.243)	0.664 (0.231)	0.580 (0.247)
LinearCCP	0.700 (0.365)	0.733 (0.195)	0.749 (0.187)	0.762 (0.174)	0.751 (0.163)	0.822 (0.300)	0.823 (0.198)	0.830 (0.211)	0.799 (0.160)	0.815 (0.161)
GPR	0.911 (0.213)	0.845 (0.230)	0.888 (0.195)	0.922 (0.126)	0.906 (0.115)	0.722 (0.304)	0.722 (0.304)	0.722 (0.304)	0.726 (0.304)	0.722 (0.304)
FANS (Ours)	1.000 (0.000)	0.986 (0.054)	0.962 (0.088)	0.968 (0.064)	0.947 (0.068)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)

graphs, underscores its potential utility in analyzing CMS in diverse network environments. Table 2 is the result from the task of dissecting the function and noise. Baselines include iSCAN, LinearCCP, and GPR (Li et al., 2023). The results from Table 2 underscore the strength of the proposed FANS method, particularly in the combined detection and dissection task on smaller graphs and in pure shift type separation on ER graphs.

We further analyze the robustness of FANS through a series of additional experiments. FANS is robust to training hyperparameters: F1 scores remain stable across batch sizes and learning rates (Appendix D.1.1), and performance degrades gracefully under reduced sample sizes (down to $N = 5,000$), shorter training (down to 100 epochs), and detection/dissection threshold in Appendix D.1.2–D.2.2. Model capacity, in contrast, has a more pronounced effect: reducing the hidden dimensions from $[32, 32, 32]$ to $[8, 8]$ causes a substantial drop in both detection and dissection, underscoring the need for sufficient CNF expressiveness to capture non-linear, non-additive mechanisms (Appendix D.2.3). We further analyze FANS under assumption violations—including quadratic-noise violations of Assumption 3.3, the ANM special case, environment swapping, and unobserved confounding in Appendix D.3.

4.2. Simultaneous shift

In Section 3.3, we propose a method to dissect function-only and simultaneous shifts. Our proposed synthetic setting already satisfies Assumption 3.3. We generate two distinct sets of shifts: (1) function-only shifts and (2) simultaneous function and noise shifts. Function-only shifts altered the causal function from $\sin(\cdot)^2$ to $\cos(2(\cdot)^2 - 3(\cdot))$. Simultaneous shifts included the same function alteration but also changed the noise distribution to $\text{Laplace}(0, 1)$. To classify a given shift, we followed the procedure outlined by Theorem 3.4.

We used a Generalized Additive Model (GAM) to estimate the conditional mean and conditional variance. Using these estimates, we

standardized the noise for each data point. The primary challenge was accurately estimating the conditional mean and variance using GAMs, resulting in moderate performance. Table 3 shows that our proposed FANS achieved a higher F1 score, indicating a superior ability

Table 3. F1-scores for distinguishing between Function-Only and Simultaneous shifts. Our method leverages Theorem 3.4.

Method	F1-score (Macro)
GPR	0.378
FANS (Ours)	0.622

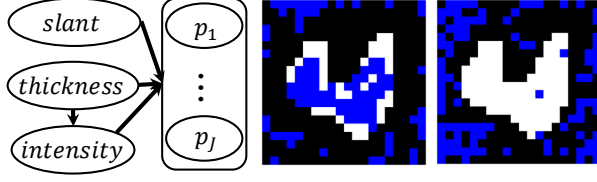


Figure 3. (Left) DAG of Morpho-MNIST, where slant, thickness, and intensity causally influence all pixel values, assuming conditional independence (Taylor-Melanson et al., 2024). (Middle) Top 100 shifted pixels (highlighted in blue) detected by FANS under the thickness function change. (Right) iSCAN results show weaker and less coherent sensitivity, confirming FANS’s superior localization of function shifts. The white pixels represent the top-100 pixels with the largest pixel-wise differences between the mean images of each environment.

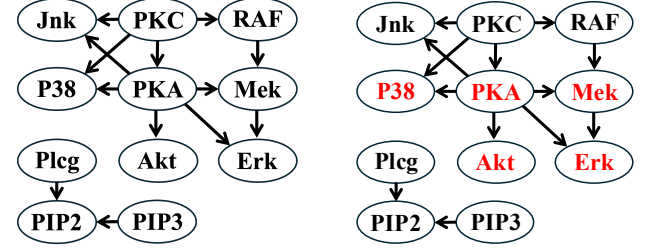
to distinguish between function-only and simultaneous shifts. The moderate F1 score reflects the inherent difficulty of this task. In Appendix D.3.1, we further examine FANS’s behavior when Assumption 3.3 is violated by quadratic noise terms, reflecting the theoretical reliance of Theorem 3.4 on the generalized affine structure.

4.3. Morpho-MNIST

We evaluated our method on the Morpho-MNIST dataset to investigate the influence of stroke thickness on pixel values, using only digit 4. We created two distinct environments by first normalizing thickness to $[-1, 1]$ and then assigning thin digits (thickness ≤ -0.5) to environment 1 and thicker digits (thickness ≥ 0) to environment 2. To simulate a function shift, we then standardize the thickness range within each environment, transforming the underlying function as $f(x) \rightarrow f((x - a)/b)$. Our method can quantify the magnitude of a detected shift. This allows us to rank all pixels based on their degree of change in Figure 3. To quantify the localization accuracy, we computed the pixel-wise difference between the empirical averages of each environment and identified the top 100 pixels with the largest ground-truth shifts. Among these, FANS correctly detected 57 pixels, whereas iSCAN only detected 3 pixels.

4.4. Real-world protein signal data

The Sachs dataset comprises flow cytometry measurements of 11 proteins and phospholipids in the human T-cell signaling pathway under various experimental perturbations. Environment 1 served as the baseline using data from general T-cell activation, while Environment 2 contained intervention data where the Mek protein was inhibited with U0126. Although a definitive ground-truth DAG for this pathway remains debated, we adopted the established network from Sachs as the underlying causal structure. Our goal was to



(a) The established causal graph of the protein signaling pathway (Kleinegesse et al., 2022)

(b) Nodes identified by FANS as having undergone a function shift are highlighted in red.

Figure 4. Application of FANS to the Sachs protein signaling dataset. The analysis identifies nodes affected by an intervention inhibiting the Mek protein (environment 2) compared to environment 1.

demonstrate which nodes experience a shift in their generative mechanisms when U0126 is introduced. Our results are consistent with the intervention targets identified by other established methods. Figure 4 visualizes these results, showing that the consensus network (Mooij et al., 2020) points to Erk as an intervention node, FCI-JCI123 identifies Mek, Erk, and PKC, while another method (Eaton & Murphy, 2007) detects RAF, Mek, Erk, Akt, PKC, and Jnk. Since an intervention on a node constitutes a function shift, our findings align with those from the Eaton and Murphy method. Notably, our method jointly identifies Mek, Erk, and Akt which is consistent with the findings of Eaton and Murphy.

5. Conclusion

We introduced FANS, a unified framework for detecting and dissecting causal mechanism shifts into function and noise components. By leveraging CNF, FANS extends dissection beyond additive noise models to general non-linear, non-additive settings, grounded in a theoretical independence criterion and a formal assumption that restores identifiability under simultaneous shifts. Empirical results on synthetic and real-world datasets corroborate our contributions.

Limitations and future work. FANS assumes a known causal DAG in the base environment and invertible structural equations, excluding discrete mechanisms and settings with unobserved confounding. The identifiability guarantee for simultaneous shifts further relies on the generalized affine interaction structure, which holds for common location-scale models but whose empirical scope merits further study. Promising directions include extending FANS to discrete variables via dequantization, integrating it with causal discovery to relax the known-DAG assumption, and leveraging its residual patterns as a diagnostic tool for hidden confounders.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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References

- Bai, J. Testing for parameter constancy in linear regressions: an empirical distribution function approach. *Econometrica: Journal of the Econometric Society*, pp. 597–622, 1996.
- Castro, D. C., Tan, J., Kainz, B., Konukoglu, E., and Glocker, B. Morpho-mnist: Quantitative assessment and diagnostics for representation learning. *Journal of Machine Learning Research*, 20(178):1–29, 2019. URL <http://jmlr.org/papers/v20/19-033.html>.
- Chang, J., Ding, Z., Jiao, Y., Li, R., and Zhijian Yang, J. Deep conditional generative learning: Model and error analysis. *arXiv e-prints*, pp. arXiv-2402, 2024.
- Chatterjee, A., Niu, Z., and Bhattacharya, B. B. A kernel-based conditional two-sample test using nearest neighbors (with applications to calibration, regression curves, and simulation-based inference). *arXiv preprint arXiv:2407.16550*, 2024.
- Chen, T., Bello, K., Aragam, B., and Ravikumar, P. iscan: identifying causal mechanism shifts among nonlinear additive noise models. *Advances in Neural Information Processing Systems*, 36:44671–44706, 2023.
- Chen, Y. and Lei, J. De-biased two-sample u-statistics with application to conditional distribution testing. *Machine Learning*, 114(2):33, 2025.
- Chow, G. C. Tests of equality between sets of coefficients in two linear regressions. *Econometrica: Journal of the Econometric Society*, pp. 591–605, 1960.
- Dash, S., Balasubramanian, V. N., and Sharma, A. Evaluating and mitigating bias in image classifiers: A causal perspective using counterfactuals. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 915–924, 2022.
- Duong, B. and Nguyen, T. Conditional independence testing via latent representation learning. In *2022 IEEE International Conference on Data Mining (ICDM)*, pp. 121–130. IEEE, 2022.
- Eaton, D. and Murphy, K. Exact bayesian structure learning from uncertain interventions. In *Artificial intelligence and statistics*, pp. 107–114. PMLR, 2007.
- Hu, X. and Lei, J. A two-sample conditional distribution test using conformal prediction and weighted rank sum. *Journal of the American Statistical Association*, 119(546): 1136–1154, 2024.
- Huang, S., Peters, J., and Pfister, N. Causal change point detection and localization. *arXiv preprint arXiv:2403.12677*, 2024.
- Javaloy, A., Sánchez-Martín, P., and Valera, I. Causal normalizing flows: from theory to practice. *Advances in Neural Information Processing Systems*, 36:58833–58864, 2023.
- Karlsson, R. and Krijthe, J. Detecting hidden confounding in observational data using multiple environments. *Advances in Neural Information Processing Systems*, 36: 44280–44309, 2023.
- Kingma, D. P., Salimans, T., Jozefowicz, R., Chen, X., Sutskever, I., and Welling, M. Improved variational inference with inverse autoregressive flow. *Advances in neural information processing systems*, 29, 2016.
- Kleinegesse, S., Lawrence, A. R., and Chockler, H. Domain knowledge in a*-based causal discovery. *arXiv preprint arXiv:2208.08247*, 2022.
- Kostic, V., Pacreau, G., Turri, G., Novelli, P., Lounici, K., and Pontil, M. Neural conditional probability for uncertainty quantification. *Advances in Neural Information Processing Systems*, 37:60999–61039, 2024.
- Lee, S., Cha, S., and Kim, I. General frameworks for conditional two-sample testing. *arXiv preprint arXiv:2410.16636*, 2024.
- Li, X., Jiang, B., and Liu, J. S. Kernel-based partial permutation test for detecting heterogeneous functional relationship. *Journal of the American Statistical Association*, 118(542):1429–1447, 2023.
- Mameche, S., Kaltenpoth, D., and Vreeken, J. Discovering invariant and changing mechanisms from data. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 1242–1252, 2022.

- Mameche, S., Vreeken, J., and Kaltenpoth, D. Identifying confounding from causal mechanism shifts. In *International Conference on Artificial Intelligence and Statistics*, pp. 4897–4905. PMLR, 2024.
- Mooij, J. M., Magliacane, S., and Claassen, T. Joint causal inference from multiple contexts. *Journal of machine learning research*, 21(99):1–108, 2020.
- Papamakarios, G., Pavlakou, T., and Murray, I. Masked autoregressive flow for density estimation. *Advances in neural information processing systems*, 30, 2017.
- Pawlowski, N., Coelho de Castro, D., and Glocker, B. Deep structural causal models for tractable counterfactual inference. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H. (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 857–869. Curran Associates, Inc., 2020.
- Pearl, J. *Causality*. Cambridge university press, 2009.
- Perron, P., Yamamoto, Y., and Zhou, J. Testing jointly for structural changes in the error variance and coefficients of a linear regression model. *Quantitative Economics*, 11(3):1019–1057, 2020.
- Perry, R., Von Kügelgen, J., and Schölkopf, B. Causal discovery in heterogeneous environments under the sparse mechanism shift hypothesis. *Advances in Neural Information Processing Systems*, 35:10904–10917, 2022.
- Pogodin, R., Schrab, A., Li, Y., Sutherland, D. J., and Gretton, A. Practical kernel tests of conditional independence. *arXiv preprint arXiv:2402.13196*, 2024.
- Ramos-Carreño, C. and Torrecilla, J. L. dcor: Distance correlation and energy statistics in Python. *SoftwareX*, 22, 2 2023. doi: 10.1016/j.softx.2023.101326. URL <https://www.sciencedirect.com/science/article/pii/S2352711023000225>.
- Reddy, A. G. and N Balasubramanian, V. Detecting and measuring confounding using causal mechanism shifts. *Advances in Neural Information Processing Systems*, 37: 61677–61699, 2024.
- Székely, G. J., Rizzo, M. L., and Bakirov, N. K. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, 2007.
- Tao, H., Yu, H., and Li, J. Deepite: Designing variational graph autoencoders for intervention target estimation. *Advances in Neural Information Processing Systems*, 37: 89978–90008, 2024.
- Taylor-Melanson, W., Sadeghi, Z., and Matwin, S. Causal generative explainers using counterfactual inference: a case study on the morpho-mnist dataset. *Pattern Analysis and Applications*, 27(3):89, 2024.
- Varici, B., Shanmugam, K., Sattigeri, P., and Tajer, A. Scalable intervention target estimation in linear models. *Advances in Neural Information Processing Systems*, 34: 1494–1505, 2021.
- Varici, B., Shanmugam, K., Sattigeri, P., and Tajer, A. Intervention target estimation in the presence of latent variables. In *Uncertainty in Artificial Intelligence*, pp. 2013–2023. PMLR, 2022.
- Yan, J., Li, Z., and Zhang, X. Distance and kernel-based measures for global and local two-sample conditional distribution testing. *arXiv preprint arXiv:2210.08149*, 2022.
- Yang, Y., Salehkaleybar, S., and Kiyavash, N. Learning unknown intervention targets in structural causal models from heterogeneous data. In *International Conference on Artificial Intelligence and Statistics*, pp. 3187–3195. PMLR, 2024.

A. Notation and Preliminaries

A.1. Notation

Variables and Metrics	
X	Random vector $X = [X_1, \dots, X_J]$, where X_j denotes the j th node
PA_j	Set of parent variables of node X_j in the DAG
ε_j	Exogenous noise variable corresponding to node X_j
k	Environment index, $k \in \{1, 2\}$
d_j	Dimension of the parent set PA_j (number of parent variables of X_j)
k_j	Output dimension of the mapping h_j
Δh_j	Functional shift defined as $h_j^{(2)} - h_j^{(1)}$
N_j	Base noise variable in CNF (e.g., standard normal distribution)
$\varepsilon^{(k)}$	Exogenous noise vector in environment k
$\tilde{\varepsilon}^{(2)}$	Inferred noise in environment 2 (shifted version of $\varepsilon^{(1)}$)
x	Observed data point in $\mathbb{R}^J$
$P_{\text{do}(\text{PA}_j=c)}^{(1)}, \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)}$	Counterfactual distributions in environments 1 and 2
$\mathcal{D}_{j,c}$	Divergence metric $\mathcal{D}(P_{\text{do}(\text{PA}_j=c)}^{(1)} \parallel \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)})$
$\bar{\mathcal{D}}_j^{\text{FANS}}$	Mean Jensen–Shannon distance over quantile conditioning points
n, J	Number of samples and number of nodes
$X_{\text{do}(\text{PA}_j=c)}^{(1)}$	Counterfactual variable in environment 1
$\tilde{X}_{\text{do}(\text{PA}_j=c)}^{(2)}$	Counterfactual variable in environment 2 (mapped via env. 1 CNF)
Functions	
f_j	A structural equation for node j which satisfies diffeomorphism
$f_j^{(k)}$	A structural equation for node j in environment k
$\mathbf{f}^{(k)}$	Collection of structural equations in environment k
$\tilde{\mathbf{f}}^{(k)}$	Non-recursive mapping obtained by unrolling the SCM in environment k such that $X^{(k)} = \tilde{\mathbf{f}}^{(k)}(\varepsilon^{(k)})$
h_j	Mapping of parent variables $\mathbb{R}^{d_j} \rightarrow \mathbb{R}^{k_j}$
$h_j^{(k)}$	Parent feature mappings in environments 1 and 2
g_j	Environment-invariant differentiable link function $\mathbb{R}^{k_j} \times \mathbb{R} \rightarrow \mathbb{R}$
q_j	Invertible mapping between true noise ε_j and base noise N_j
$q(\cdot)$	True invertible function mapping SCM noises to CNF base noises
$\hat{q}$	Estimated invertible transformation of $q(\cdot)$
$\hat{T}^{(1)}, (\hat{T}^{(1)})^{-1}$	Learned CNF and inverse CNF for environment 1
$\mathcal{I}(x, j, c)$	Function computing the modified noise vector under $\text{do}(\text{PA}_j = c)$

A.2. Preliminaries

Definition A.1 (Causal Normalizing Flow). (Javaloy et al., 2023) Causal Normalizing Flow (CNF) $T(X)$ is an autoregressive normalizing flow (ANF) (Kingma et al., 2016; Papamakarios et al., 2017) to learn an invertible transformation from i.i.d random variables $X = [X_1, \dots, X_J]$ to a set of independent base-noise variables $N = [N_1, \dots, N_J]$.

$$T(X) = T(\tilde{\mathbf{f}}(\varepsilon)) = [N_1, \dots, N_J]. \quad (14)$$

Base noise variables N are random variables that follow simple base distributions, such as the standard normal distribution. A CNF learns a mapping from observed data X to these base noise variables N . Once the CNF is trained, we can map X to N or vice versa due to its invertibility. Theorem 1 in (Javaloy et al., 2023) states that if a true SCM and a CNF match, meaning both data generating processes yield the same distribution, the difference between the two mechanisms lies in an invertible function q_j . Specifically, q_j takes the true exogenous noise of the SCM as input and outputs the corresponding CNF base noise variable N_j .

$$q_j(\varepsilon_j) = N_j, \quad \forall j \in \{1, \dots, J\} \quad (15)$$

B. PROOFS

B.1. Counterfactual distribution in CNF

Let $\varepsilon^{(1)} \sim P_{\varepsilon^{(1)}}$, $\tilde{\varepsilon}^{(2)} \sim P_{\tilde{\varepsilon}^{(2)}}$. Let $\mathcal{I}(x, j, c)$ be the function that computes the modified noise vector required to perform $\text{do}(\text{PA}_j = c)$ on an initial data point x . We denote the parent indices as j^* . The procedure is defined as:

1. Infer initial noise: $\hat{u} = \hat{T}^{(1)}(x)$.
2. Create an intervened data vector by substitution: $\text{pa}_j \leftarrow c = (x_1, \dots, c, \dots, x_J)$, where the parent of j th node is replaced by c .
3. Recalculate the noise component: $u_{\text{PA}_j \leftarrow c} = \left(\hat{T}^{(1)}(\text{pa}_j \leftarrow c) \right)_{j^*}$.
4. Assemble the final modified noise vector: $\hat{u}_{\text{do}(\text{PA}_j=c)} = (\hat{u}_1, \dots, u_{\text{PA}_j \leftarrow c}, \dots, \hat{u}_J)$.

Thus, $\mathcal{I}(x, j, c) = \hat{u}_{\text{do}(\text{PA}_j=c)}$. The counterfactual random variable is defined as

$$X_{\text{do}(\text{PA}_j=c)}^{(1)} = (\hat{T}^{(1)})^{-1}(\mathcal{I}(X^{(1)}, j, c)), \quad \text{where } X_{\text{do}(\text{PA}_j=c)}^{(1)} \sim P_{\text{do}(\text{PA}_j=c)}^{(1)}.$$

Similarly,

$$\tilde{X}_{\text{do}(\text{PA}_j=c)}^{(2)} = (\hat{T}^{(1)})^{-1}(\mathcal{I}(X^{(2)}, j, c)), \quad \text{where } \tilde{X}_{\text{do}(\text{PA}_j=c)}^{(2)} \sim \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)}.$$

The distribution of $\tilde{X}_{\text{do}(\text{PA}_j=c)}^{(2)}$ is denoted as $\tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)}$. The test statistic for detecting a causal mechanism shift is defined as

$$\mathcal{D}_{j,c} := \mathcal{D}(P_{\text{do}(\text{PA}_j=c)}^{(1)} \parallel \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)}).$$

B.2. Proof of Theorem 3.1

Theorem 3.1 A causal mechanism shift at node X_j has occurred if and only if the expected divergence is strictly greater than zero.

$$P(X_j^{(1)} \mid \text{PA}_j^{(1)}) \neq P(X_j^{(2)} \mid \text{PA}_j^{(2)}) \iff \mathbb{E}_{c \sim P(\text{PA}_j^{(1)})}[\mathcal{D}_{j,c}] > 0,$$

where $\mathcal{D}_{j,c} := \mathcal{D}(P_{\text{do}(\text{PA}_j=c)}^{(1)} \parallel \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)})$ and the counterfactual laws are defined via the CNF map $\mathcal{I}$ as below.

Interventional pushforward defined via CNF. Let $\hat{T}^{(1)} : \mathbb{R}^J \rightarrow \mathbb{R}^J$ be the learned invertible CNF for Environment 1 with inverse $(\hat{T}^{(1)})^{-1}$, and write

$$U^{(1)} = \hat{T}^{(1)}(X^{(1)}) \sim P_{U^{(1)}}, \quad \hat{U}^{(2)} = \hat{T}^{(1)}(X^{(2)}) \sim P_{\hat{U}^{(2)}}.$$

Given x , target j , and value c , define the modified-noise map

$$\mathcal{I}(x, j, c) = (\hat{u}_1, \dots, \hat{u}_{j-1}, (\hat{T}^{(1)}(\text{pa}_j \leftarrow c))_{j^*}, \hat{u}_{j+1}, \dots, \hat{u}_J), \quad \hat{u} = \hat{T}^{(1)}(x).$$

Set $G_{j,c} := (\hat{T}^{(1)})^{-1} \circ \mathcal{I}(\cdot, j, c)$. Then

$$X_{\text{do}(\text{PA}_j=c)}^{(1)} := G_{j,c}(X^{(1)}) \sim P_{\text{do}(\text{PA}_j=c)}^{(1)}, \quad \tilde{X}_{\text{do}(\text{PA}_j=c)}^{(2)} := G_{j,c}(X^{(2)}) \sim \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)}.$$

Assume absolute continuity of all laws and that $\mathcal{D}$ is nonnegative and equals 0 iff its arguments coincide.

Lemma (Jacobian cancels) Let $x_{inv} = G_{j,c}^{-1}(x)$ and $u_{inv} = \hat{T}^{(1)}(x_{inv})$. Then,

$$\tilde{p}_{\text{do}(\text{PA}_j=c)}^{(2)}(x) = p_{\text{do}(\text{PA}_j=c)}^{(1)}(x) \cdot \mathcal{C}(x), \quad \mathcal{C}(x) := \frac{p_{\hat{U}^{(2)}}(u_{inv})}{p_{U^{(1)}}(u_{inv})}.$$

Proof of lemma. By change of variables under the common diffeomorphism $G_{j,c}$, and setting $x_{inv} = G_{j,c}^{-1}(x)$:

$$\begin{aligned} p_{\text{do}(\text{PA}_j=c)}^{(1)}(x) &= p_{X^{(1)}}(G_{j,c}^{-1}(x)) \left| \det J_{G_{j,c}^{-1}}(x) \right| = p_{X^{(1)}}(x_{inv}) \left| \det J_{G_{j,c}^{-1}}(x_{inv}) \right|, \\ \tilde{p}_{\text{do}(\text{PA}_j=c)}^{(2)}(x) &= p_{X^{(2)}}(G_{j,c}^{-1}(x)) \left| \det J_{G_{j,c}^{-1}}(x) \right| = p_{X^{(2)}}(x_{inv}) \left| \det J_{G_{j,c}^{-1}}(x_{inv}) \right|. \end{aligned}$$

The ratio of the densities is $\mathcal{C}(x) = \frac{p_{X^{(2)}}(x_{inv})}{p_{X^{(1)}}(x_{inv})}$, as the Jacobian $\left| \det J_{G_{j,c}^{-1}}(x) \right|$ cancels. Now, we apply the change of variables for the CNF map $\hat{T}^{(1)}$ at the point x_{inv} , using $u_{inv} = \hat{T}^{(1)}(x_{inv})$:

$$\begin{aligned} p_{X^{(1)}}(x_{inv}) &= p_{U^{(1)}}(\hat{T}^{(1)}(x_{inv})) \left| \det J_{\hat{T}^{(1)}}(x_{inv}) \right| = p_{U^{(1)}}(u_{inv}) \left| \det J_{\hat{T}^{(1)}}(u_{inv}) \right|, \\ p_{X^{(2)}}(x_{inv}) &= p_{\hat{U}^{(2)}}(\hat{T}^{(1)}(x_{inv})) \left| \det J_{\hat{T}^{(1)}}(x_{inv}) \right| = p_{\hat{U}^{(2)}}(u_{inv}) \left| \det J_{\hat{T}^{(1)}}(u_{inv}) \right|. \end{aligned}$$

Substituting these into the ratio for $\mathcal{C}(x)$, the second Jacobian $\left| \det J_{\hat{T}^{(1)}}(x_{inv}) \right|$ also cancels, yielding $\mathcal{C}(x) = \frac{p_{\hat{U}^{(2)}}(u_{inv})}{p_{U^{(1)}}(u_{inv})}$. $\triangle$

Forward direction ($\implies$). Suppose a mechanism shift at X_j occurs, i.e., $P(X_j^{(1)} \mid \text{PA}_j^{(1)}) \neq P(X_j^{(2)} \mid \text{PA}_j^{(2)})$. For the shifted node j , $P_{\hat{U}_j^{(2)}} \neq P_{U_j^{(1)}}$. Consequently, the joint noise laws differ, $P_{\hat{U}^{(2)}} \neq P_{U^{(1)}}$, meaning $p_{\hat{U}^{(2)}}(u) \neq p_{U^{(1)}}(u)$ on a set of positive measure. By the lemma, $\mathcal{C}(x) = \frac{p_{\hat{U}^{(2)}}(u_{inv})}{p_{U^{(1)}}(u_{inv})}$. Since $u_{inv} = (\hat{T}^{(1)} \circ G_{j,c}^{-1})(x)$ is a diffeomorphism, u_{inv} maps sets of positive measure to sets of positive measure. Thus, $\mathcal{C}(x) \neq 1$ on a set of positive measure. This implies the interventional densities differ, $\tilde{p}_{\text{do}(\text{PA}_j=c)}^{(2)} \neq p_{\text{do}(\text{PA}_j=c)}^{(1)}$, so $\mathcal{D}_{j,c} > 0$. Therefore

$$\mathbb{E}_{c \sim P(\text{PA}_j^{(1)})}[\mathcal{D}_{j,c}] \geq \int_A \mathcal{D}_{j,c} dP(\text{PA}_j^{(1)} = c) > 0.$$

Converse direction ($\impliedby$). Assume $\mathbb{E}_c[\mathcal{D}_{j,c}] > 0$. Then the set $B := \{c : \mathcal{D}_{j,c} > 0\}$ has positive $P(\text{PA}_j^{(1)})$ -measure. For any $c \in B$, $\mathcal{D}_{j,c} > 0$ implies $p_{\text{do}(\text{PA}_j=c)}^{(1)} \neq \tilde{p}_{\text{do}(\text{PA}_j=c)}^{(2)}$. By the lemma, this means $\mathcal{C}(x) = \frac{p_{\hat{U}^{(2)}}(u_{inv})}{p_{U^{(1)}}(u_{inv})} \neq 1$. Since u_{inv} covers the support of the noise distributions, this implies $p_{\hat{U}^{(2)}}(\cdot) \neq p_{U^{(1)}}(\cdot)$ as functions, and thus $P_{\hat{U}^{(2)}} \neq P_{U^{(1)}}$. Assuming modularity ($P_{\hat{U}_k^{(2)}} = P_{U_k^{(1)}}$ for $k \neq j$), the fact $P_{\hat{U}^{(2)}} \neq P_{U^{(1)}}$ necessarily implies that the j -th noise components must differ: $P_{\hat{U}_j^{(2)}} \neq P_{U_j^{(1)}}$. This corresponds to a mechanism shift at node j : $P(X_j^{(1)} \mid \text{PA}_j^{(1)}) \neq P(X_j^{(2)} \mid \text{PA}_j^{(2)})$.

B.3. Proof of Theorem 3.2

Theorem 3.2 For a shifted node X_j , the statistical independence of its inferred noise $\tilde{\varepsilon}_j^{(2)}$ and its parents $\text{PA}_j^{(2)}$ indicates the type of shift:

$$\begin{aligned} \tilde{\varepsilon}_j^{(2)} \not\perp \text{PA}_j^{(2)} &\iff \text{the shift is a function shift.} \\ \tilde{\varepsilon}_j^{(2)} \perp \text{PA}_j^{(2)} &\iff \text{the shift is a noise shift.} \end{aligned}$$

We infer the noises in environment 2 using the SCM from environment 1: $[\tilde{\varepsilon}_1^{(2)}, \dots, \tilde{\varepsilon}_J^{(2)}] = \tilde{f}^{-1}(X_1^{(2)}, \dots, X_J^{(2)})$. We denote $\tilde{\varepsilon}_j^{(2)}$ as the inferred noise of environment 2 data based on $\mathcal{M}^{(1)}$.

$$\begin{aligned} X_j^{(1)} &= g_j(h_j^{(1)}(\text{PA}_j^{(1)}), \varepsilon_j^{(1)}) \\ X_j^{(2)} &= g_j(h_j^{(2)}(\text{PA}_j^{(2)}), \varepsilon_j^{(2)}) = g_j(h_j^{(1)}(\text{PA}_j^{(2)}), \tilde{\varepsilon}_j^{(2)}). \end{aligned}$$

$\varepsilon_j^{(2)}$ and $\tilde{\varepsilon}_j^{(2)}$ lie in the same probability space, while $\varepsilon_j^{(1)}$ and $\varepsilon_j^{(2)}$ lie in different probability spaces. Let $(\Omega^{(2)}, \mathcal{F}^{(2)}, \mathbb{P}^{(2)})$ be the underlying probability space of environment 2, and let

$$\varepsilon_j^{(2)} : \Omega^{(2)} \rightarrow \mathbb{R}$$

be random variables on this space such that

$$X_j^{(2)} = g_j(h_j^{(2)}(\text{PA}_j^{(2)}), \varepsilon_j^{(2)}) \quad \text{a.s.}$$

By Assumption 2.2, for every fixed $p \in \mathbb{R}^{d_j}$ the map

$$\varepsilon \mapsto g_j(h_j^{(1)}(p), \varepsilon)$$

is bijective, so its inverse in the second argument exists. We therefore define the residual noise $\tilde{\varepsilon}_j^{(2)} : \Omega^{(2)} \rightarrow \mathbb{R}$ by

$$\tilde{\varepsilon}_j^{(2)}(\omega) := (g_j(h_j^{(1)}(\text{PA}_j^{(2)}(\omega)), \cdot))^{-1}(X_j^{(2)}(\omega)), \quad \omega \in \Omega^{(2)}.$$

Then, by construction,

$$X_j^{(2)}(\omega) = g_j(h_j^{(1)}(\text{PA}_j^{(2)}(\omega)), \tilde{\varepsilon}_j^{(2)}(\omega)) \quad \text{for all } \omega \in \Omega^{(2)},$$

so both $\varepsilon_j^{(2)}$ and $\tilde{\varepsilon}_j^{(2)}$ are random variables on the same probability space $(\Omega^{(2)}, \mathcal{F}^{(2)}, \mathbb{P}^{(2)})$.

UNDER FUNCTION SHIFT

We begin by defining the function shift Δh_j as the deviation between the mechanism functions in the two environments:

$$\Delta h_j(\text{PA}_j^{(2)}) := h_j^{(2)}(\text{PA}_j^{(2)}) - h_j^{(1)}(\text{PA}_j^{(2)}). \quad (16)$$

By equating the observed variable $X_j^{(2)}$ generated from the true mechanism with its reconstruction using the inferred noise $\tilde{\varepsilon}_j^{(2)}$ and the base mechanism, we establish the following identity:

$$g_j(h_j^{(1)}(\text{PA}_j^{(2)}) + \Delta h_j(\text{PA}_j^{(2)}), \varepsilon_j^{(2)}) = g_j(h_j^{(1)}(\text{PA}_j^{(2)}), \tilde{\varepsilon}_j^{(2)}).$$

To rigorously analyze the dependence of the noise difference $\eta^* := \tilde{\varepsilon}_j^{(2)} - \varepsilon_j^{(2)}$ on the parents, we define the implicit function F as follows:

$$F(\eta, \mathbf{pa}; \varepsilon_j^{(2)}) := g_j(h_j^{(1)}(\mathbf{pa}) + \Delta h_j(\mathbf{pa}), \varepsilon_j^{(2)}) - g_j(h_j^{(1)}(\mathbf{pa}), \varepsilon_j^{(2)} + \eta). \quad (17)$$

Note that $F(\eta^*, \mathbf{pa}; \varepsilon_j^{(2)}) = 0$. Under Assumption 2.2, since g_j is differentiable and invertible with respect to the noise term, its partial derivative is non-zero:

$$\frac{\partial F}{\partial \eta}(\eta^*, \mathbf{pa}; \varepsilon_j^{(2)}) \neq 0. \quad (18)$$

Thus, by the Implicit Function Theorem, there exists a function Φ such that $\eta^* = \Phi(\mathbf{pa}, \varepsilon_j^{(2)})$. Since $\Delta h_j \notin \mathcal{T}_{g_j}$, $\Phi(\mathbf{pa}, \varepsilon_j^{(2)})$ cannot be reduced to a function of $\varepsilon_j^{(2)}$ alone independent of $\mathbf{pa}$. Crucially, Φ depends on $\mathbf{pa}$ because F involves $\mathbf{pa}$ through both the original mechanism $h_j^{(1)}$ and the shift Δh_j .

Finally, since the true noise is independent of parents ($\varepsilon_j^{(2)} \perp \text{PA}_j^{(2)}$) while the difference term depends on them ($\eta^* = \Phi(\mathbf{pa}, \varepsilon_j^{(2)})$), the inferred noise $\tilde{\varepsilon}_j^{(2)} = \varepsilon_j^{(2)} + \Phi(\text{PA}_j^{(2)}, \varepsilon_j^{(2)})$ must be dependent on the parents. Therefore, we conclude:

$$\tilde{\varepsilon}_j^{(2)} \not\perp \text{PA}_j^{(2)}.$$

For example, we assume linear functions with Gaussian noises.

$$\begin{aligned} X_j^{(1)} &= \beta_j^{(1)} \mathbf{PA}_j^{(1)} + \varepsilon_j^{(1)}, \varepsilon_j^{(1)} \sim \mathcal{N}(0, s^2) \\ X_j^{(2)} &= \beta_j^{(2)} \mathbf{PA}_j^{(2)} + \varepsilon_j^{(2)}, \varepsilon_j^{(2)} \sim \mathcal{N}(0, s^2) \\ &= \beta_j^{(1)} \mathbf{PA}_j^{(2)} + \tilde{\varepsilon}_j^{(2)}. \end{aligned}$$

Then, $\tilde{\varepsilon}_j^{(2)} \sim \mathcal{N}((\beta_j^{(2)} - \beta_j^{(1)})\mathbf{PA}_j^{(2)}, s^2)$. Since $\tilde{\varepsilon}_j^{(2)}$ and $\varepsilon_j^{(2)}$ lie in the same probability space, $\eta^* := \tilde{\varepsilon}_j^{(2)} - \varepsilon_j^{(2)} = (\beta_j^{(2)} - \beta_j^{(1)})\mathbf{PA}_j^{(2)}$.

UNDER NOISE SHIFT

Under a pure noise shift at node X_j , we assume that the structural functions are unchanged across environments, $h_j^{(2)} = h_j^{(1)}$.

$$X_j^{(2)} = g_j(h_j^{(2)}(\mathbf{PA}_j^{(2)}), \varepsilon_j^{(2)}) = g_j(h_j^{(1)}(\mathbf{PA}_j^{(2)}), \varepsilon_j^{(2)}).$$

By the definition of the residual noise $\tilde{\varepsilon}_j^{(2)}$,

$$X_j^{(2)} = g_j(h_j^{(1)}(\mathbf{PA}_j^{(2)}), \tilde{\varepsilon}_j^{(2)})$$

For each fixed parent value $\mathbf{pa}$, the map $\varepsilon \mapsto g_j(h_j^{(1)}(\mathbf{pa}), \varepsilon)$ is invertible by Assumption 2.2. Thus, it concludes that $\tilde{\varepsilon}_j^{(2)} = \varepsilon_j^{(2)}$ which is independent of parent variables.

B.4. Proof of Theorem 3.4

Our strategy is that under a function-only shift if $q_j(\tilde{\varepsilon}_j^{(2)}) = \tilde{N}_j^{(2)}$ is expressed as a linear function of $q_j(\varepsilon_j^{(2)}) = N_j^{(2)}$, then we can standardize and square it to get a chi-square distribution. However, when a function and noise simultaneously shift, then this process doesn't ensure chi-square distribution since it induces an extra randomness. We let $b_{j,0}(\mathbf{pa})$ and $b_{j,1}(\mathbf{pa})$ are arbitrary functions of $\mathbf{pa}$. Then, under a function-only shift, we assume $\tilde{N}_j^{(2)}$ is expressed as a following linear expression:

$$\tilde{N}_j^{(2)} = N_j^{(2)} b_{j,1}(\mathbf{pa}) + b_{j,0}(\mathbf{pa}).$$

Since $N_j^{(2)} \perp \mathbf{PA}_j^{(2)}$ and $E[N_j^{(2)}] = 0$,

$$E[\tilde{N}_j^{(2)} | \mathbf{PA}_j^{(2)}] = E[N_j^{(2)} b_{j,1}(\mathbf{PA}_j^{(2)}) + b_{j,0}(\mathbf{PA}_j^{(2)}) | \mathbf{PA}_j^{(2)}] = b_{j,0}(\mathbf{PA}_j^{(2)}).$$

Also, since $\text{Var}(N_j^{(2)}) = 1$,

$$\text{Var}(\tilde{N}_j^{(2)} | \mathbf{PA}_j^{(2)}) = (b_{j,1}(\mathbf{PA}_j^{(2)}))^2 \text{Var}(N_j^{(2)}) = (b_{j,1}(\mathbf{PA}_j^{(2)}))^2.$$

Now, consider the test statistics $\tilde{q}_j(\tilde{\varepsilon}_j^{(2)})$ defined as follows:

$$\tilde{q}_j(\tilde{\varepsilon}_j^{(2)}) := \frac{\tilde{N}_j^{(2)} - E[\tilde{N}_j^{(2)} | \mathbf{PA}_j^{(2)}]}{\sqrt{\text{Var}(\tilde{N}_j^{(2)} | \mathbf{PA}_j^{(2)})}} = \frac{b_{j,1}(\mathbf{PA}_j^{(2)})}{|b_{j,1}(\mathbf{PA}_j^{(2)})|} N_j^{(2)} = \text{sgn}(b_{j,1}(\mathbf{PA}_j^{(2)})) N_j^{(2)}.$$

To handle the sign function, we square the standardized noise.

$$(\tilde{q}_j(\tilde{\varepsilon}_j^{(2)}))^2 = (\text{sgn}(b_{j,1}(\mathbf{PA}_j^{(2)})) N_j^{(2)})^2 = (N_j^{(2)})^2.$$

Assumption 3.3 describes

$$g_j^*(h_j, N_j) = \Psi_j(\beta_{j,1}(h_j) \cdot N_j + \beta_{j,0}(h_j)).$$

When we consider a function-only shift in environment 2,

$$\begin{aligned} X_j^{(2)} &= g_j^*(h_j^{(2)}, N_j^{(2)}) = \Psi_j(\beta_{j,1}(h_j^{(2)}) \cdot N_j^{(2)} + \beta_{j,0}(h_j^{(2)})) \\ &= g_j^*(h_j^{(1)}, \tilde{N}_j^{(2)}) = \Psi_j(\beta_{j,1}(h_j^{(1)}) \cdot \tilde{N}_j^{(2)} + \beta_{j,0}(h_j^{(1)})). \end{aligned}$$

Since Ψ_j is invertible,

$$\beta_{j,1}(h_j^{(2)}) \cdot N_j^{(2)} + \beta_{j,0}(h_j^{(2)}) = \beta_{j,1}(h_j^{(1)}) \cdot \tilde{N}_j^{(2)} + \beta_{j,0}(h_j^{(1)}).$$

Then,

$$\tilde{N}_j^{(2)} = N_j^{(2)} b_{j,1}(\mathbf{pa}_j) + b_{j,0}(\mathbf{pa}_j).$$

The converse does not hold in general. If the shift in $N_j^{(2)}$ is limited to location and scale, the square of the standardized variable ($\tilde{N}_j^{(2)}$) follows $\chi^2(1)$. However, in the case of a function-noise shift affecting higher order moments (i.e., third moment or higher), the standardized distribution deviates from normality. Consequently, it implies

$$(\tilde{q}_j(\tilde{\varepsilon}_j^{(2)}))^2 \not\sim \chi^2(1).$$

B.5. Proof of Proposition 3.7 and Corollary 3.8

In this appendix, we provide a proof of Proposition 3.7 and Corollary 3.8.

B.5.1. SETUP AND NOTATION

Fix a node j and an environment $k \in \{1, 2\}$. We write $N := N_j^{(k)}$ for the true residual noise, $\hat{N} := \hat{N}_j^{(k)} = \hat{q}_j(\tilde{\varepsilon}_j^{(k)})$ for its CNF-based estimate, and $Z := Z^{(k)} = \text{PA}_j^{(k)}$ for the parents. Let $(N', \hat{N}', Z')$ and $(N'', \hat{N}'', Z'')$ denote independent copies sharing the joint distribution of $(N, \hat{N}, Z)$.

The squared distance covariance of random vectors U, V admits the expectation form (Székely et al., 2007)

$$\begin{aligned} \mathcal{V}^2(U, V) &= \mathbb{E}[\|U - U'\| \cdot \|V - V'\|] + \mathbb{E}[\|U - U'\|] \cdot \mathbb{E}[\|V - V'\|] \\ &\quad - 2 \mathbb{E}[\|U - U'\| \cdot \|V - V''\|], \end{aligned} \tag{19}$$

and the distance correlation is defined as

$$\mathcal{R}(U, V) := \begin{cases} \sqrt{\frac{\mathcal{V}^2(U, V)}{\sqrt{\mathcal{V}^2(U, U)} \cdot \sqrt{\mathcal{V}^2(V, V)}}} & \text{if } \mathcal{V}^2(U, U) \cdot \mathcal{V}^2(V, V) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{20}$$

Under finite second moments, $0 \leq \mathcal{R}(U, V) \leq 1$, with equality to zero if and only if U and V are independent.

Population-level analysis. Throughout this appendix, $\mathcal{V}^2(U, V)$ and $\mathcal{R}(U, V)$ denote the *population* distance covariance and distance correlation, defined with respect to the joint distribution of (U, V) . Our goal is to quantify how the CNF approximation error δ_n propagates to the test statistic at the population level. In practice, these quantities are replaced by their empirical V-statistic estimators computed from finite test samples, which introduces an additional $O_p(1/\sqrt{m})$ sampling error by standard results (Székely et al., 2007). This sampling error is orthogonal to the CNF-induced error analyzed here and does not affect the leading-order behavior as both $n, m \rightarrow \infty$.

Moreover, all O_p and o_p statements below are with respect to the randomness of CNF training (i.e., the sample used to estimate $\hat{q}_j$), which is captured by the rate δ_n in Assumption 3.5.

The analysis relies on two conditions. Assumption 3.5 (L^2 CNF consistency) is interpreted here as follows: there exists a deterministic sequence $\delta_n \rightarrow 0$ as the CNF training sample size $n \rightarrow \infty$ such that, with probability one over CNF training,

$$\mathbb{E}_{X^{(k)}}[(\hat{N}_j^{(k)} - N_j^{(k)})^2 \mid \hat{q}_j] \leq \delta_n^2, \quad (21)$$

where the conditional expectation is taken over the data distribution of environment k given the trained CNF map $\hat{q}_j$. Assumption 3.6 (finite second moments) ensures that $\mathbb{E}[(N_j^{(k)})^2] < \infty$ and $\mathbb{E}[\|Z^{(k)}\|^2] < \infty$ for each j and k .

Throughout, define $C_Z := \sqrt{\mathbb{E}[\|Z - Z'\|^2]}$ and $C_N := \sqrt{\mathbb{E}[(N - N')^2]}$, both finite by Assumption 3.6. All bounds in Lemmas B.1 and B.2 hold conditionally on a CNF realization satisfying Assumption 3.5; by the L^2 consistency, such realizations occur with probability one in the limit.

B.5.2. PROOF

Lemma B.1 (Distance Covariance Error Bound). *Under Assumptions 3.5 and 3.6,*

$$|\mathcal{V}^2(\hat{N}, Z) - \mathcal{V}^2(N, Z)| \leq 8 C_Z \delta_n. \quad (22)$$

Proof. By the triangle inequality applied to the three terms in the definition (19),

$$|\mathcal{V}^2(\hat{N}, Z) - \mathcal{V}^2(N, Z)| \leq |A_1| + |A_2| + 2|A_3|, \quad (23)$$

where

$$A_1 := \mathbb{E}[|\hat{N} - \hat{N}'| \|Z - Z'\|] - \mathbb{E}[|N - N'| \|Z - Z'\|], \quad (24)$$

$$A_2 := (\mathbb{E}[|\hat{N} - \hat{N}'|] - \mathbb{E}[|N - N'|]) \cdot \mathbb{E}[\|Z - Z'\|], \quad (25)$$

$$A_3 := \mathbb{E}[|\hat{N} - \hat{N}'| \|Z - Z''\|] - \mathbb{E}[|N - N'| \|Z - Z''\|]. \quad (26)$$

Bound on $|A_1|$. By the reverse triangle inequality $|\hat{N} - \hat{N}'| - |N - N'| \leq |\hat{N} - N| + |\hat{N}' - N'|$,

$$|A_1| \leq \mathbb{E}[|\hat{N} - N| \|Z - Z'\|] + \mathbb{E}[|\hat{N}' - N'| \|Z - Z'\|]. \quad (27)$$

Applying Cauchy–Schwarz to each term,

$$\mathbb{E}[|\hat{N} - N| \|Z - Z'\|] \leq \sqrt{\mathbb{E}[(\hat{N} - N)^2]} \cdot \sqrt{\mathbb{E}[\|Z - Z'\|^2]} \leq \delta_n C_Z, \quad (28)$$

where the last inequality invokes Assumption 3.5 and the definition of C_Z . By the identical distribution of $(\hat{N}, N)$ and $(\hat{N}', N')$, the second term has the same bound. Therefore

$$|A_1| \leq 2\delta_n C_Z. \quad (29)$$

Bound on $|A_2|$. Since $\mathbb{E}[\|Z - Z'\|] \leq \sqrt{\mathbb{E}[\|Z - Z'\|^2]} = C_Z$ by Jensen’s inequality,

$$\begin{aligned} |A_2| &\leq |\mathbb{E}[|\hat{N} - \hat{N}'|] - \mathbb{E}[|N - N'|]| \cdot C_Z \\ &\leq \mathbb{E}[|\hat{N} - \hat{N}'| - |N - N'|] \cdot C_Z \\ &\leq (\mathbb{E}[|\hat{N} - N|] + \mathbb{E}[|\hat{N}' - N'|]) \cdot C_Z \\ &\leq 2 \sqrt{\mathbb{E}[(\hat{N} - N)^2]} \cdot C_Z \leq 2\delta_n C_Z, \end{aligned} \quad (30)$$

where the third inequality is the reverse triangle inequality and the penultimate inequality is Jensen’s applied to $|\cdot|$.

Bound on $|A_3|$. Replacing Z' with Z'' in the argument for A_1 , and using the identical joint distribution of (Z, Z'') and (Z, Z') (since Z'' is another i.i.d. copy),

$$|A_3| \leq 2\delta_n C_Z. \quad (31)$$

Combining (29)–(31),

$$|\mathcal{V}^2(\hat{N}, Z) - \mathcal{V}^2(N, Z)| \leq 2\delta_n C_Z + 2\delta_n C_Z + 2 \cdot 2\delta_n C_Z = 8\delta_n C_Z. \quad (32)$$

□

Lemma B.2 (Distance Variance Error Bound). *Under Assumptions 3.5 and 3.6,*

$$|\mathcal{V}^2(\hat{N}, \hat{N}) - \mathcal{V}^2(N, N)| \leq 16 C_N \delta_n + o(\delta_n). \quad (33)$$

Proof. Setting $V = U$ in (19) and applying the triangle inequality,

$$|\mathcal{V}^2(\hat{N}, \hat{N}) - \mathcal{V}^2(N, N)| \leq |B_1| + |B_2| + 2|B_3|, \quad (34)$$

where

$$B_1 := \mathbb{E}[|\hat{N} - \hat{N}'|^2] - \mathbb{E}[|N - N'|^2], \quad (35)$$

$$B_2 := (\mathbb{E}[|\hat{N} - \hat{N}'|])^2 - (\mathbb{E}[|N - N'|])^2, \quad (36)$$

$$B_3 := \mathbb{E}[|\hat{N} - \hat{N}'| \cdot |\hat{N} - \hat{N}''|] - \mathbb{E}[|N - N'| \cdot |N - N''|]. \quad (37)$$

The key observation is that each term now involves two instances of $\hat{N}$ (versus two of N), so the reverse triangle inequality accumulates twice as many error contributions compared to Lemma B.1.

Bound on $|B_1|$. Factoring the difference of squares,

$$\begin{aligned} |B_1| &\leq \mathbb{E}[(|\hat{N} - \hat{N}'| + |N - N'|) \cdot (|\hat{N} - \hat{N}'| - |N - N'|)] \\ &\leq \mathbb{E}[(|\hat{N} - \hat{N}'| + |N - N'|)(|\hat{N} - N| + |\hat{N}' - N'|)]. \end{aligned} \quad (38)$$

By Cauchy–Schwarz,

$$\begin{aligned} |B_1| &\leq \sqrt{\mathbb{E}[(|\hat{N} - \hat{N}'| + |N - N'|)^2]} \cdot \sqrt{\mathbb{E}[(|\hat{N} - N| + |\hat{N}' - N'|)^2]} \\ &\leq \sqrt{2\mathbb{E}[(\hat{N} - \hat{N}')^2] + 2\mathbb{E}[(N - N')^2]} \cdot \sqrt{4\delta_n^2} \\ &\leq 2\delta_n \cdot (2C_N + o(1)) = 4\delta_n C_N + o(\delta_n), \end{aligned} \quad (39)$$

where the final step uses $\mathbb{E}[(\hat{N} - \hat{N}')^2] \rightarrow \mathbb{E}[(N - N')^2] = C_N^2$ as $n \rightarrow \infty$, a direct consequence of Assumption 3.5.

Bound on $|B_2|$. Using $a^2 - b^2 = (a + b)(a - b)$,

$$\begin{aligned} |B_2| &= |\mathbb{E}[|\hat{N} - \hat{N}'|] + \mathbb{E}[|N - N'|]| \cdot |\mathbb{E}[|\hat{N} - \hat{N}'|] - \mathbb{E}[|N - N'|]| \\ &\leq (2C_N + o(1)) \cdot 2\delta_n = 4\delta_n C_N + o(\delta_n), \end{aligned} \quad (40)$$

where the second inequality applies the same reverse-triangle-plus-Jensen argument as in (30) to the second factor, and Jensen’s inequality ($\mathbb{E}[|\cdot|] \leq \sqrt{\mathbb{E}[|\cdot|^2]}$) together with Assumption 3.5 to the first factor.

Bound on $|B_3|$. By the reverse triangle inequality applied to the product $|\hat{N} - \hat{N}'| \cdot |\hat{N} - \hat{N}''|$,

$$\begin{aligned} |B_3| &\leq \mathbb{E}[|\hat{N} - N|(|\hat{N} - \hat{N}''| + |N - N''|)] + \mathbb{E}[|\hat{N}' - N'| \cdot |\hat{N} - \hat{N}''|] \\ &\quad + \mathbb{E}[|N - N'| \cdot |\hat{N} - N|] + \mathbb{E}[|N - N'| \cdot |\hat{N}'' - N''|]. \end{aligned} \quad (41)$$

Each of the four terms is bounded by Cauchy–Schwarz and Assumption 3.5, giving $|B_3| \leq 4\delta_n C_N + o(\delta_n)$.

Summing the three contributions,

$$|B_1| + |B_2| + 2|B_3| \leq 4\delta_n C_N + 4\delta_n C_N + 2 \cdot 4\delta_n C_N + o(\delta_n) = 16\delta_n C_N + o(\delta_n). \quad (42)$$

□

Lemma B.2 implies in particular that $\mathcal{V}^2(\hat{N}, \hat{N}) \xrightarrow{p} \mathcal{V}^2(N, N)$ as $n \rightarrow \infty$.

Lemma B.3 (Denominator Stability). *Under Assumptions 3.5 and 3.6, and assuming $\mathcal{V}^2(N, N) > 0$ and $\mathcal{V}^2(Z, Z) > 0$, define*

$$D(U, V) := \sqrt{\mathcal{V}^2(U, U) \cdot \mathcal{V}^2(V, V)}. \quad (43)$$

Then $D(\hat{N}, Z) \xrightarrow{p} D(N, Z) > 0$ as $n \rightarrow \infty$, and moreover $|D(\hat{N}, Z) - D(N, Z)| = O_p(\delta_n)$.

Proof. By Lemma B.2, $\mathcal{V}^2(\hat{N}, \hat{N}) = \mathcal{V}^2(N, N) + O_p(\delta_n)$, and $\mathcal{V}^2(Z, Z)$ is deterministic and positive. Applying the mean value theorem to $u \mapsto \sqrt{u}$ on a neighborhood of $\mathcal{V}^2(N, N) > 0$ yields $|\sqrt{\mathcal{V}^2(\hat{N}, \hat{N})} - \sqrt{\mathcal{V}^2(N, N)}| = O_p(\delta_n)$. Multiplying by the constant $\sqrt{\mathcal{V}^2(Z, Z)}$ gives the stated rate. $\square$

The non-degeneracy conditions $\mathcal{V}^2(N, N), \mathcal{V}^2(Z, Z) > 0$ hold whenever N and Z are non-trivial (not almost surely constant), which is the setting of interest for dissection.

Lemma B.4 (Rate of Distance Correlation Convergence). *Under Assumptions 3.5 and 3.6, and assuming non-degeneracy of N and Z ,*

$$|\mathcal{R}(\hat{N}, Z) - \mathcal{R}(N, Z)| = O_p(\sqrt{\delta_n}). \quad (44)$$

Moreover, if $\mathcal{R}(N, Z) > 0$, the rate sharpens to $O_p(\delta_n)$.

Proof. Let $r := \mathcal{V}^2(N, Z)/D(N, Z)$ and $\hat{r} := \mathcal{V}^2(\hat{N}, Z)/D(\hat{N}, Z)$, so that $\mathcal{R}(N, Z) = \sqrt{r}$ and $\mathcal{R}(\hat{N}, Z) = \sqrt{\hat{r}}$.

We first bound $|\hat{r} - r|$. Writing

$$\hat{r} - r = \frac{\mathcal{V}^2(\hat{N}, Z) - \mathcal{V}^2(N, Z)}{D(\hat{N}, Z)} + \mathcal{V}^2(N, Z) \cdot \frac{D(N, Z) - D(\hat{N}, Z)}{D(\hat{N}, Z) D(N, Z)}, \quad (45)$$

and applying Lemmas B.1 and B.3,

$$|\hat{r} - r| = O_p(\delta_n), \quad (46)$$

since both numerators are $O_p(\delta_n)$ and the denominators converge in probability to positive constants.

Case 1: $r = 0$ ($\mathcal{R}(N, Z) = 0$). Since $\hat{r} \geq 0$ almost surely ($\mathcal{V}^2 \geq 0$) and $\hat{r} = |\hat{r} - r| = O_p(\delta_n)$ by (46),

$$\mathcal{R}(\hat{N}, Z) = \sqrt{\hat{r}} = O_p(\sqrt{\delta_n}). \quad (47)$$

Case 2: $r > 0$ ($\mathcal{R}(N, Z) > 0$). Using the identity $\sqrt{\hat{r}} - \sqrt{r} = \frac{\hat{r} - r}{\sqrt{\hat{r}} + \sqrt{r}}$ and noting that $\sqrt{\hat{r}} + \sqrt{r} \xrightarrow{p} 2\sqrt{r} > 0$, we obtain

$$|\sqrt{\hat{r}} - \sqrt{r}| = \frac{|\hat{r} - r|}{\sqrt{\hat{r}} + \sqrt{r}} = O_p(\delta_n). \quad (48)$$

Uniform bound. Since $\delta_n \leq \sqrt{\delta_n}$ for $\delta_n \leq 1$ (which holds for n sufficiently large as $\delta_n \rightarrow 0$), the rate $O_p(\sqrt{\delta_n})$ subsumes both cases, establishing (44). $\square$

Proof of Proposition 3.7. Applying Lemma B.4 separately to environments $k = 1$ and $k = 2$,

$$\hat{\mathcal{R}}_1 := \mathcal{R}(\hat{N}^{(1)}, Z^{(1)}) \xrightarrow{p} \mathcal{R}(N^{(1)}, Z^{(1)}) =: \mathcal{R}_1, \quad (49)$$

$$\hat{\mathcal{R}}_2 := \mathcal{R}(\hat{N}^{(2)}, Z^{(2)}) \xrightarrow{p} \mathcal{R}(N^{(2)}, Z^{(2)}) =: \mathcal{R}_2. \quad (50)$$

Componentwise convergence in probability implies joint convergence, so by the continuous mapping theorem applied to $(a, b) \mapsto |a - b|$,

$$\Delta \hat{\mathcal{R}} = |\hat{\mathcal{R}}_2 - \hat{\mathcal{R}}_1| \xrightarrow{p} |\mathcal{R}_2 - \mathcal{R}_1| = \Delta \mathcal{R}^*. \quad (51)$$

We further characterize the rate of convergence:

- **Pure noise shift** ($\mathcal{R}_1 = \mathcal{R}_2 = 0$, so $\Delta\mathcal{R}^* = 0$): Case 1 of Lemma B.4 applies to both environments, giving $\Delta\hat{\mathcal{R}} = O_p(\sqrt{\delta_n})$.
- **Function shift** ($\mathcal{R}_2 > 0$, so $\Delta\mathcal{R}^* > 0$): Case 2 applies to environment 2, so $|\hat{\mathcal{R}}_2 - \mathcal{R}_2| = O_p(\delta_n)$, while $\hat{\mathcal{R}}_1 = O_p(\sqrt{\delta_n})$ from Case 1 (since under the null we have $\mathcal{R}_1 = 0$). Combining via the triangle inequality yields $|\Delta\hat{\mathcal{R}} - \Delta\mathcal{R}^*| = O_p(\sqrt{\delta_n})$.

In both regimes, $\Delta\hat{\mathcal{R}} \xrightarrow{p} \Delta\mathcal{R}^*$ as $n \rightarrow \infty$. □

Proof of Corollary 3.8. We verify the two claims for any fixed rejection threshold $\tau_{\text{dis}} > 0$.

Type I error (pure noise shift, $\Delta\mathcal{R}^ = 0$).* By Proposition 3.7, $\Delta\hat{\mathcal{R}} \xrightarrow{p} 0$. Hence

$$\Pr(\Delta\hat{\mathcal{R}} > \tau_{\text{dis}}) = \Pr(|\Delta\hat{\mathcal{R}} - 0| > \tau_{\text{dis}}) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (52)$$

Type II error (function shift, $\Delta\mathcal{R}^ > 0$).* Fix $\tau_{\text{dis}} \in (0, \Delta\mathcal{R}^*)$ and let $\epsilon := \Delta\mathcal{R}^* - \tau_{\text{dis}} > 0$. Since

$$\{\Delta\hat{\mathcal{R}} \leq \tau_{\text{dis}}\} = \{\Delta\mathcal{R}^* - \Delta\hat{\mathcal{R}} \geq \epsilon\} \subseteq \{|\Delta\hat{\mathcal{R}} - \Delta\mathcal{R}^*| \geq \epsilon\}, \quad (53)$$

we have

$$\Pr(\Delta\hat{\mathcal{R}} \leq \tau_{\text{dis}}) \leq \Pr(|\Delta\hat{\mathcal{R}} - \Delta\mathcal{R}^*| \geq \epsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (54)$$

by Proposition 3.7. Equivalently, $\Pr(\Delta\hat{\mathcal{R}} > \tau_{\text{dis}}) \rightarrow 1$. □

C. Experiment Details

For all synthetic experiments, we generate 50,000 samples per environment. Of these, 40,000 are used to train the CNF, and the remaining 10,000 are held out as a validation set.

We denote S^{method} as a set of shift detected nodes by a method, and $S^{\text{method,func}}$ is a set of function shift detected nodes by a method and $S^{\text{method,noise}}$ is a set of noise shift detected nodes. This section explains how these sets are derived from each methodology in detail. Furthermore, S^{true} is a true set of shifted nodes, which is identical to S .

C.1. Synthetic dataset for detection and dissection

C.1.1. FANS

We implement FANS on top of the Causal Normalizing Flow (CNF) architecture of Javaloy et al. (2023). We use the following default configuration across all synthetic experiments: a Neural Spline Flow as the flow layer, hidden layer dimensions of $[32, 32, 32]$, batch size of 512, learning rate of 0.01 with decay, and 3000 training epochs. The Adam optimizer is used throughout. A sensitivity analysis of these hyperparameters is provided in Appendix D.1.1.

All scores are computed over the set of nodes that have at least one parent. For detection of shift, the two conditional densities are estimated by kernel density estimation on a shared grid, and $\mathcal{D}_{j,c}$ is evaluated as the square root of the Jensen–Shannon divergence (JSD):

$$\mathcal{D}_{j,c} := \sqrt{\text{JSD}(P_{\text{do}(\text{PA}_j=c)}^{(1)}, \tilde{P}_{\text{do}(\text{PA}_j=c)}^{(2)})}.$$

We approximate the overall distributional shift by evaluating $\mathcal{D}_{j,c}$ at three conditioning points $c \in \{c^1, c^2, c^3\}$, selected as the 0.25, 0.5, and 0.75 quantiles of the empirical distribution of PA_j in environment 1. For a multi-dimensional parent vector PA_j , these quantile points are computed element-wise. We then define the aggregate measure $\bar{\mathcal{D}}_j^{\text{FANS}}$ as the average of $\mathcal{D}_{j,c}$ over these three points, and identify shifted nodes by a fixed threshold τ :

$$S^{\text{FANS}} := \{j : \bar{\mathcal{D}}_j^{\text{FANS}} > \tau\}, \quad \tau = 0.1.$$

Since the detection stage only requires a cut-off on the sorted divergence, any data-driven selection rule is directly applicable. In particular, an elbow (knee) criterion locating the elbow index e of the sorted profile, as used by iSCAN (Chen et al., 2023), is a natural way when no a priori scale for the divergence is available. We use the fixed value in the experiments.

Table 4. Detection F1 under the fixed threshold $\tau = 0.1$ versus the threshold-free elbow rule, over 30 DAGs per setting. Standard deviations in parentheses.

Detection rule	ER Graph – Nodes (J)					SF Graph – Nodes (J)				
	10	20	30	40	50	10	20	30	40	50
Fixed $\tau = 0.1$	0.989 (0.061)	1.000 (0.000)	0.980 (0.043)	0.982 (0.030)	0.975 (0.037)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
Elbow rule	0.978 (0.085)	0.996 (0.020)	0.992 (0.023)	0.982 (0.032)	0.981 (0.073)	1.000 (0.000)	0.989 (0.034)	0.987 (0.029)	0.965 (0.068)	0.926 (0.151)

As an ablation, we re-run the detection stage with the elbow (knee) rule of iSCAN (Chen et al., 2023) in place of the fixed threshold, leaving everything else unchanged.

Furthermore, for the node index $j \in S^{\text{true}}$, we perform a dissection to determine whether the shift is primarily due to a function shift or a noise shift. For each node index j in S^{true} , we derive the inferred noise $q(\hat{\varepsilon}_j^{(k)})$ for both environments $k \in \{1, 2\}$. Let $\mathcal{R}(\cdot, \cdot)$ denote the empirical distance correlation. We compute the difference between the distance correlation in Environment 2 and the baseline correlation in Environment 1. If this difference exceeds a predefined threshold τ_{dis} (e.g., 0.05), we determine that the independence condition is violated. Consequently, j is classified as a function shift; otherwise, it is classified as a noise shift.

$$\begin{aligned}\rho_j &:= \mathcal{R}\left(q(\hat{\varepsilon}_j^{(2)}), \text{PA}_j^{(2)}\right) - \mathcal{R}\left(q(\hat{\varepsilon}_j^{(1)}), \text{PA}_j^{(1)}\right) \\ S^{\text{FANS,func}} &:= \{j \in S^{\text{true}} : \rho_j > \tau_{\text{dis}}\} \\ S^{\text{FANS,noise}} &:= S^{\text{true}} \setminus S^{\text{FANS,func}}.\end{aligned}$$

The consistency analysis in Section 3.4 is stated for the absolute difference, which is the symmetric form required by the continuous mapping argument. The implementation uses the signed difference, because a function shift usually increases the residual dependence relative to the environment 1 baseline. Other methods such as permutation tests based on distance covariance (Ramos-Carreño & Torrecilla, 2023) can be applied.

C.1.2. iSCAN

For the initial shift detection task, iSCAN returns a score s_j^{iSCAN} for each node j , which serves as our initial test statistic for shift detection. A higher score indicates a higher likelihood of a shift. The null hypothesis $H_{0,j}^{\text{iSCAN}} : s_j^{\text{iSCAN}} < t$ (node j is not shifted) is rejected if $s_j \geq t$. The threshold t was derived by elbow strategy in the original iSCAN paper (Chen et al., 2023). The set of nodes identified as shifted by iSCAN is

$$S^{\text{iSCAN}} := \{j : s_j^{\text{iSCAN}} \geq t\}.$$

For the dissection task, concerning nodes $j \in S^{\text{iSCAN}}$, we implement Algorithm 5 in (Chen et al., 2023) to detect functionally shifted edges. We model the functional relationship as nonlinear function between a node X_j and its parents PA_j which has a following relationship.

$$X_j^{(k)} = f_j^{(k)}(\text{PA}_j^{(k)}) + \varepsilon_j^{(k)}, \quad k \in \{1, 2\}.$$

Following the iSCAN methodology, $f_j^{(k)} : \mathbb{R}^{d_j} \mapsto \mathbb{R}$ is assumed to have an additive form $f_j^{(k)}(\text{PA}_j^{(k)}) = \sum_{p=1}^{d_j} f_{j,p}^{(k)}([\text{PA}_j^{(k)}]_p)$, where $f_{j,p}^{(k)} : \mathbb{R} \mapsto \mathbb{R}$ are univariate functions. We approximate each $f_{j,p}^{(k)}$ using a B-spline expansion. Specifically, we use cubic B-spline with 4 degrees of freedom for each parent component $[\text{PA}_j^{(k)}]_p$. The i -th basis function is denoted as $B_{j,i,p}(x_{j,p}^{(k)})$ for a node j and p -th parent variable. We denote $\mathbf{B}_{j,p}^{(k)} = [B_{j,1,p}(x_{j,p}^{(k)}), \dots, B_{j,4,p}(x_{j,p}^{(k)})]^T$ be the column vector of these basis functions. Similarly, $\tilde{\beta}_{j,p} = [\tilde{\beta}_{j,1,p}^{(k)}, \dots, \tilde{\beta}_{j,4,p}^{(k)}]^T$ be the column vector of corresponding

coefficients for $p \in \{1, \dots, d_j\}$. The approximation is given by:

$$f_j^{(k)}(\text{PA}_j^{(k)}) \approx \tilde{\beta}_{j,0}^{(k)} + \sum_{p=1}^{d_j} (\tilde{\beta}_{j,p}^{(k)})^T \mathbf{B}_{j,p}^{(k)}([\text{PA}_j^{(k)}]_p),$$

where $\tilde{\beta}_{j,0}^{(k)}$ is the intercept. Since our setting assumes a given DAG, we use the known parent set without estimating them. Let $\tilde{\beta}_j^{(k)}$ denote the vector concatenating all coefficient vectors $\beta_{j,p}^{(k)}$ (for $p = 1, \dots, d_j$).

A function shift is characterized by a change in these estimated B-spline coefficients. The null hypothesis of no functional shift for node $j \in S^{\text{iSCAN}}$ is:

$$H_{0,j}^{\text{iSCAN,func}} : \tilde{\beta}_j^{(1)} = \tilde{\beta}_j^{(2)}.$$

This hypothesis is tested using Hotelling's T-squared test with a significance level of 0.05. A rejection of $H_{0,j}^{\text{iSCAN,func}}$ indicates a functional shift. The set of functionally shifted nodes is:

$$S^{\text{iSCAN,func}} := \{j \in S^{\text{true}} : H_{0,j}^{\text{iSCAN,func}} \text{ is rejected at the 0.05 significance level}\}.$$

If a node $j \in S^{\text{iSCAN}}$ is identified as shifted by the initial scored-based criterion but is not found to have a functional shift, it is then classified as a noise shift. Thus, the set of noise-shifted nodes is:

$$S^{\text{iSCAN,noise}} := S^{\text{true}} \setminus S^{\text{iSCAN,func}}.$$

C.1.3. LINEARCCP

LinearCCP (Huang et al., 2024) was originally designed for scenarios where information about parent sets is unavailable. However, as our experimental setting assumes a given DAG, we adapt LCCP by utilizing the known parent sets for each node j as independent variables in a linear causal model. For each environment $k \in \{1, 2\}$, the model for node j is:

$$X_j^{(k)} = \sum_{p=1}^{d_j} \beta_{j,p}^{(k)} [\text{PA}_j^{(k)}]_p + \varepsilon_j^{(k)},$$

where $\beta_{j,p}^{(k)}$ are the corresponding linear coefficients. Let $\beta_j^{(k)} = [\beta_{j,1}^{(k)}, \dots, \beta_{j,d_j}^{(k)}]^T$ be the vector of these coefficients. These coefficients $\hat{\beta}_j^{(k)}$ are estimated using Ordinary Least Squares (OLS), and the corresponding residuals are denoted as $\hat{\varepsilon}_j^{(k)}$. LCCP is used to detect two types of shifts: functional shifts (changes in coefficients) and noise shifts (change in the error distribution).

The two null hypotheses of no functional shift or no noise shift for node index j are:

$$\begin{aligned} H_{0,j}^{\text{LCCP,func}} : \beta_j^{(1)} &= \beta_j^{(2)} \\ H_{0,j}^{\text{LCCP,noise}} : F_{\varepsilon_j^{(1)}}(x) &= F_{\varepsilon_j^{(2)}}(x) \text{ for all } x. \end{aligned}$$

$H_{0,j}^{\text{LCCP,func}}$ is tested using the Chow test (Chow, 1960). If the p-value from this test is less than the significance level of 0.05, the null hypothesis is rejected, indicating a functional shift. Also, $H_{0,j}^{\text{LCCP,noise}}$ is tested using a two-sample Kolmogorov-Smirnov (KS) test on the estimated residuals $\hat{\varepsilon}_j^{(1)}$ and $\hat{\varepsilon}_j^{(2)}$ at a 0.05 significance level. If any of these hypotheses is rejected, we classify the node index j as being shifted.

A dissection task is executed following other methods like iSCAN and FANS. If $H_{0,j}^{\text{LCCP,func}}$ is rejected, we decide j is a function shift detected node. Noise shifted nodes are then determined by difference of S^{true} and $S^{\text{LCCP,func}}$

$$\begin{aligned} S^{\text{LCCP,func}} &:= \{j \in S^{\text{true}} | (H_{0,j}^{\text{LCCP,func}} \text{ is rejected})\} \\ S^{\text{LCCP,noise}} &:= S^{\text{true}} \setminus S^{\text{LCCP,func}}. \end{aligned}$$

C.1.4. SPLITKCI

We implement the Split Kernel Conditional Independence (SplitKCI) test (Pogodin et al., 2024) to detect shifted nodes between two environments. The test evaluates the null hypothesis:

$$H_{0,j}^{\text{SplitKCI}} : X_j \perp Z \mid \text{PA}_j,$$

for each node j , where $Z \in \{1, 2\}$ denotes the environment indicator and PA_j are all the parent variables of node j in the known DAG structure. The SplitKCI test is a kernel-based conditional independence test with bias reduction via data splitting.

Although SplitKCI is formulated as a conditional independence test of the form $X_j \perp Z \mid \text{PA}_j$, it effectively functions as a *two-sample conditional test*. Specifically, it tests whether the conditional distribution $P(X_j \mid \text{PA}_j)$ remains invariant across two different environments. This is achieved by embedding samples from both environments into reproducing kernel Hilbert spaces (RKHS), and measuring the dependence between the target variable X_j and the environment label Z , conditioning on PA_j . Thus, SplitKCI operates the two-sample conditional hypothesis by testing for violations of node shifts in a nonparametric manner.

Specifically, we evaluate a conditional independence statistic using the Hilbert-Schmidt norm of a cross-covariance operator:

$$T_j^{\text{SplitKCI}} := \left\langle C_j^{(1)}, C_j^{(2)} \right\rangle_{\text{HS}},$$

where $C_j^{(1)}$ and $C_j^{(2)}$ are Hilbert-Schmidt operators constructed from two independent splits of the data. These operators are computed using conditional mean embeddings (CME) estimated via kernel ridge regression from PA_j to X_j , and from PA_j to Z . To control for bias in finite samples, the CME is estimated on two disjoint subsets of the data. This reduces dependence between estimators and enhances validity of the test. We use Gaussian kernels for both X_j and PA_j . The significance of each T_j^{SplitKCI} is assessed via the wild bootstrap method. We draw Rademacher weights w_i and compute the test statistic under bootstrapped samples to form a null distribution. The p-value for each node is estimated as:

$$p_j = \frac{1}{B} \sum_{b=1}^B \mathbb{I} \left(T_{j,b}^{\text{boot}} \geq T_j^{\text{SplitKCI}} \right),$$

where B is the number of bootstrap replicates. The node j is declared shifted if $p_j < \alpha$ for a significance level $\alpha = 0.05$. The set of shifted nodes is defined as:

$$S^{\text{SplitKCI}} := \{j : p_j < 0.05\}.$$

C.1.5. PREITER

We evaluate the PreDITer algorithm proposed by Varici et al. (2022), which detects shifted nodes between two environments. PreDITer exploits the fact that soft interventions induce sparse changes in the precision matrix of a linear SEM, thereby formulating the task as a *precision difference estimation* problem.

Specifically, PreDITer models the observed precision matrix in each environment k as $\Theta^{(k)} = (\Sigma^{(k)})^{-1}$, where $\Sigma^{(k)}$ is the covariance matrix of observed variables under environment k . For a pair of environments (1, 2), the difference $\Delta = \Theta^{(1)} - \Theta^{(2)}$ reveals the set of potentially affected nodes $S_\Delta = \{i : [\Delta]_{ii} \neq 0\}$. A node i is identified as intervened if no subset $S \subseteq S_\Delta$ yields an invariant diagonal element $[\Delta_S]_{ii} = 0$, according to Lemma 2 of Varici et al. (2022). Similarly, parent and spouse sets of each intervened node are recovered via nonzero off-diagonal entries $[\Delta_S]_{ij} \neq 0$.

The method employs the precision-difference estimation (PDE) based on an ADMM solver with ℓ_1 -penalty:

$$\Delta^* = \arg \min_{\Delta} \frac{1}{2} \text{tr}(\Delta^\top \Sigma^{(1)} \Delta \Sigma^{(2)}) - \text{tr}(\Delta(\Sigma^{(1)} - \Sigma^{(2)})) + \lambda \|\Delta\|_1.$$

The resulting sparse estimate Δ^* is symmetrized and thresholded to determine the effective intervention targets K . The overall set of detected shifted nodes is defined as

$$S^{\text{PreDITer}} := \{j : \|\Delta_{j,\cdot}^*\|_1 > 0\}.$$

C.1.6. GPR

Gaussian Process Regression (GPR)-based Partial Permutation Test (Li et al., 2023) determines whether a shift in the conditional relationship between a node X_j and its parent variables PA_j is attributable to a functional shift or a noise shift. Under our implemented framework, we assume that a set of true shifted nodes S^{True} has already been given. GPR is used as a follow-up test for classifying the nature of the shift. For each node $j \in S^{\text{True}}$, we model the structural equation in each environment $k \in \{1, 2\}$ as:

$$X_j^{(k)} = f_j^{(k)}(\text{PA}_j^{(k)}) + \varepsilon_j^{(k)},$$

where $f_j^{(k)}$ is a smooth (possibly nonlinear) function, and $\varepsilon_j^{(k)}$ is zero-mean Gaussian noise. We define four hypotheses modeling the relationship $X_j^{(k)} = f_j^{(k)}(\text{PA}_j^{(k)}) + \varepsilon_j^{(k)}$ across environments $k \in \{1, 2\}$:

1. **Null Model (H_0)** Assumes a shared function and shared noise variance across environments ($f^{(1)} = f^{(2)}$ and $\sigma_{(1)}^2 = \sigma_{(2)}^2$).
2. **Noise Shift Model (H_{noise})** Assumes a shared function but allows environment-specific noise variances ($f^{(1)} = f^{(2)}$ and $\sigma_{(1)}^2 \neq \sigma_{(2)}^2$).
3. **Function Shift Model (H_{func})** Assumes environment-specific functions but a shared noise variance ($f^{(1)} \neq f^{(2)}$ and $\sigma_{(1)}^2 = \sigma_{(2)}^2$).
4. **Simultaneous Shift Model (H_{simul})** Assumes both environment-specific functions and environment-specific noise variances ($f^{(1)} \neq f^{(2)}$ and $\sigma_{(1)}^2 \neq \sigma_{(2)}^2$).

We compute the maximum log-likelihood estimates (MLE) for the data under each hypothesis, denoted as ℓ_0 , ℓ_{noise} , ℓ_{func} , and ℓ_{simul} , respectively.

Dissection (Function vs. Noise) To distinguish between function shift and noise shift, we compare the likelihood improvements of H_{func} and H_{noise} over the null model H_0 . We define the likelihood ratios as $\Delta_{\text{func}} = \ell_{\text{func}} - \ell_0$ and $\Delta_{\text{noise}} = \ell_{\text{noise}} - \ell_0$. More precisely, the classification rule is:

$$\text{Shift Type}_j = \begin{cases} \text{Function} & \text{if } \Delta_{\text{func}} > \Delta_{\text{noise}}, \\ \text{Noise} & \text{otherwise.} \end{cases}$$

That is, if separating functions yields a greater likelihood gain than allowing for variance heterogeneity, the shift is primarily functional.

C.2. Simultaneous Shift

C.2.1. FANS

To distinguish between function-only shifts and simultaneous shifts (where both function and noise distributions change), we employ a statistical testing procedure grounded in Theorem 3.4 of the main paper. The theorem posits that under a function-only shift (and satisfying Assumption 3.3), the squared standardized inferred noise must converge to a Chi-squared distribution with one degree of freedom, $\chi^2(1)$. Deviations from this theoretical distribution imply the presence of a simultaneous shift.

We implement this test using a non-parametric approach to estimate the conditional moments required for standardization. This procedure avoids strong parametric assumptions about the functional form of the shift and consists of the following three steps:

1. Conditional Moment Estimation via GAMs We utilize Generalized Additive Models (GAMs) to estimate the conditional mean and variance of the inferred noise $N_j^{(2)} = \hat{q}_j(\tilde{\varepsilon}_j^{(2)})$ given the parent variables $\text{PA}_j^{(2)}$. First, we fit a GAM with spline terms to estimate the conditional expectation $\hat{\mu}_j(pa) \approx \mathbb{E}[N_j^{(2)} \mid \text{PA}_j^{(2)} = pa]$. To capture complex dependencies and interactions between multiple parent variables, we include tensor product terms for pairwise interactions in the GAM formulation. Subsequently, we compute the squared residuals $r^2 = (N_j^{(2)} - \hat{\mu}_j(\text{PA}_j^{(2)}))^2$ and fit a second

GAM to estimate the conditional variance $\hat{\sigma}_j^2(pa) \approx \text{Var}(N_j^{(2)} \mid PA_j^{(2)} = pa)$. We enforce a minimum variance floor to ensure numerical stability during the subsequent standardization step.

2. Noise Standardization Using the estimated conditional moments, we compute the standardized noise Z_j for each data point in Environment 2 as follows:

$$Z_j = \frac{N_j^{(2)} - \hat{\mu}_j(PA_j^{(2)})}{\sqrt{\hat{\sigma}_j^2(PA_j^{(2)})}} \quad (55)$$

Under the null hypothesis of a function-only shift, the square of this standardized variable, Z_j^2 , follows a $\chi^2(1)$ distribution.

3. Hypothesis Testing Finally, we perform a Kolmogorov-Smirnov (KS) test to compare the empirical distribution of the squared standardized residuals Z_j^2 against the theoretical cumulative distribution function of $\chi^2(1)$.

- **Function-Only Shift:** If the p-value of the KS test is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis. We conclude that the standardized noise is consistent with $\chi^2(1)$, indicating the shift is fully explained by the functional change.
- **Simultaneous Shift:** If the p-value is less than $\alpha = 0.05$, we reject the null hypothesis. This indicates that the standardized noise deviates from the expected theoretical distribution, suggesting that the noise distribution itself has also shifted (Simultaneous Shift).

C.2.2. GPR

To identify simultaneous shifts, we compare the function-only shift model (H_{func}) against the simultaneous shift model (H_{simul}). We evaluate whether adding noise heterogeneity to a function shift model significantly improves the fit. The decision rule is based on the comparison of Δ_{func} and $\Delta_{\text{simul}} = \ell_{\text{simul}} - \ell_0$:

$$\text{Shift Type}_j = \begin{cases} \text{Simultaneous} & \text{if } \Delta_{\text{simul}} > \Delta_{\text{func}}, \\ \text{Function} & \text{otherwise.} \end{cases}$$

That is, if the simultaneous model provides a higher likelihood gain, we can conclude that both the function and the noise distribution have been shifted.

C.3. Morpho-MNIST

The Morpho-MNIST dataset extends the MNIST digits with quantitative morphometric attributes—thickness, length, width, height, and slant—to enable systematic analysis of representation learning (Castro et al., 2019). Building on its causal extensions, such as DeepSCM (Pawlowski et al., 2020) and ImageCFGen (Dash et al., 2022), we adopt a similar causal framework but modify it to focus exclusively on the digit ‘4’. By excluding the digit label (l) from the generative process, we isolate the effects of continuous morphological variables—thickness (t), intensity (i), and slant (s)—to examine their causal interactions within a controlled single-digit setting.

We define a causal graph over the image by treating each pixel as a node and assuming that all morphological attributes causally affect every pixel. We assume that the pixel-level nodes do not have causal relationships with one another. We crop the image borders to obtain a 20×20 resolution. We separate environment 1 and environment 2 based on the thickness attribute. We first min-max normalize thickness values to the range $[-1, 1]$, then assign samples with normalized thickness ≤ -0.5 to environment 1 and those with thickness ≥ 0 to environment 2. This preprocessing results in 2,645 images in environment 1 and 370 images in environment 2.

We use the normalized thickness for environment separation; all morphological variables are normalized using the mean and standard deviation of samples within each environment. By doing so, a same thickness value has a different meaning in different environments, leading to node shifts. For the image data, we preprocess the pixel nodes so that their values, originally ranging from 0 to 255 in the grayscale MNIST images, are centered around 127.5 and rescaled to the range $[-1, 1]$. We construct the adjacency matrix by ordering nodes according to the causal graph, placing root nodes first. Thus, we define the node order as $s, t, i, p_0, \dots, p_{399}$.

FANS trains a normalizing flow on env1 ($n = 2645$, internal 80/20 train/validation split), then computes per-node Jensen–Shannon distance on an evaluation pair of 370 samples drawn from each environment. For iSCAN, a single statistical two-sample test (no training phase) is applied to a pair of 370 samples drawn from each environment.

C.4. Sachs protein signaling dataset

The Sachs dataset consists of flow cytometry measurements of 11 proteins and phospholipids in a human T-cell signaling pathway under multiple experimental perturbations.

Each variable is log-transformed and then standardized to zero mean and unit variance within each environment separately. This removes pure marginal location and scale differences between the two conditions, so the detected shifts reflect changes in the conditional mechanisms.

Shift detection criterion (Sachs). For node-wise shift detection, we use the same detection-stage Jensen–Shannon statistic $\bar{D}_j$ and the same fixed threshold $\tau = 0.1$ as in the synthetic experiments.

C.5. Computations

All experiments were conducted on a workstation equipped with an Intel(R) Core(TM) i9-10900X CPU (10 cores) and 4× NVIDIA GeForce RTX 4090 GPUs. We measured the wall-clock time for each method, representing the total duration required to process one configuration (consisting of 30 datasets). While parallel processing was utilized where feasible, GPR was executed serially to maintain efficiency, as processing a single node with GPR consumes over 70% of the entire CPU capacity. SplitKCI was also executed serially for efficiency. LinearCCP performs detection and dissection in a single pass, so the same measurement appears in both tables.

Table 5. Runtime comparison for Shift Detection task on ER and SF graphs across varying node counts.

Method	10 Nodes	20 Nodes	30 Nodes	40 Nodes	50 Nodes
<i>ER graph</i>					
iSCAN	17s	29s	42s	55s	1m 9s
SplitKCI	4m 20s	8m 37s	12m 20s	16m 35s	20m 22s
LCIT	3m 43s	7m 15s	10m 6s	13m 30s	17m 16s
LinearCCP	1s	1s	2s	3s	3s
PreDITer	<1s	<1s	1s	3s	7m 10s
FANS (Inference)	6s	13s	18s	33s	43s
FANS (Train)	189m 30s	190m 32s	193m 48s	195m 54s	197m 24s
<i>SF graph</i>					
iSCAN	16s	29s	41s	55s	1m 9s
SplitKCI	6m 28s	13m 37s	21m 0s	28m 19s	35m 38s
LCIT	7m 2s	14m 43s	23m 2s	30m 46s	39m 32s
LinearCCP	<1s	<1s	1s	1s	1s
PreDITer	<1s	<1s	1s	10s	10s
FANS (Inference)	8s	24s	44s	1m 2s	1m 4s
FANS (Train)	189m 39s	191m 25s	194m 35s	196m 1s	197m 38s

D. Additional Analyses and Ablations

D.1. Training Hyperparameter and Threshold Sensitivity

D.1.1. HYPERPARAMETER SENSITIVITY

We additionally report the loss of the trained CNF, evaluated at each run’s best-validation checkpoint and averaged over 30 random datasets per variant in Table 7. We used this quantity to select hyperparameters for the synthetic experiments. Note that the log-likelihood only measures how well the CNF fits the observational data and is therefore not a direct evaluation of how well each option performs at detection or dissection.

Table 6. Runtime comparison for Shift Dissection task on ER and SF graphs across varying node counts.

Method	10 Nodes	20 Nodes	30 Nodes	40 Nodes	50 Nodes
<i>ER graph</i>					
iSCAN	<1s	<1s	1s	1s	1s
LinearCCP	1s	1s	2s	3s	3s
GPR	97m 39s	285m 14s	314m 48s	557m 30s	486m 51s
FANS (Inference)	<1s	1s	1s	1s	2s
<i>SF graph</i>					
iSCAN	<1s	<1s	<1s	<1s	<1s
LinearCCP	<1s	<1s	1s	1s	1s
GPR	90m 19s	252m 2s	298m 54s	359m 56s	407m 6s
FANS (Inference)	<1s	<1s	<1s	<1s	<1s

Table 8 demonstrates that FANS is robust to hyperparameter choices of batch size and learning rate: F1 scores vary marginally across 8 configurations.

 Table 7. Validation loss across flow architectures and training hyperparameters. Each cell reports mean validation loss over 30 DAGs. Lower is better. The best configuration is highlighted in **bold**.

Flow	LR	Batch Size			
		512	1024	2048	4096
NSF	0.01	56.02	56.09	56.15	56.28
	0.001	56.58	56.80	56.99	57.17
NAF	0.01	57.27	57.15	57.23	57.35
	0.001	57.34	57.70	58.08	58.60

Table 8. Hyperparameter sensitivity for FANS across flow architectures (NSF, NAF) on 50-node ER graphs. Each cell reports F1 (Detection) / F1 (Dissection). Hidden layers fixed at [32,32,32], 3000 epochs. The default configuration is highlighted in bold.

Flow	LR	Batch Size			
		512	1024	2048	4096
NSF	0.01	0.975 / 0.947	0.975 / 0.945	0.971 / 0.951	0.976 / 0.940
	0.001	0.964 / 0.935	0.964 / 0.928	0.966 / 0.936	0.962 / 0.921
NAF	0.01	0.942 / 0.883	0.928 / 0.891	0.932 / 0.882	0.920 / 0.899
	0.001	0.925 / 0.879	0.920 / 0.882	0.916 / 0.872	0.908 / 0.872

D.1.2. DETECTION AND DISSECTION THRESHOLDS

Table 9 reports F1 scores on 50-node ER graphs across a range of plausible values. FANS relies on two thresholds: (i) a divergence threshold τ used in the detection stage to classify a node as shifted if its aggregated divergence $\bar{\mathcal{D}}_j$ exceeds τ , and (ii) a distance correlation threshold used in the dissection stage to classify a shifted node as a function shift versus a noise shift. We examine the sensitivity of FANS to manual choices of both thresholds.

D.2. Robustness to Sample Size and Model Capacity

D.2.1. SAMPLE SIZE

We evaluate how FANS performance scales with training sample size. Using the 50-node ER graph setting with the default configuration, we reduce the number of samples per environment from 50,000 to 5,000, holding out 20% of the data for

Table 9. Threshold sensitivity on 50-node ER graphs.

Task	Threshold value	F1 Score
Detection (τ)	0.06	0.965
	0.08	0.995
	0.10	0.975
	0.12	0.921
	0.14	0.835
Dissection	0.02	0.937
	0.04	0.941
	0.06	0.935
	0.08	0.906

validation in each case.

Table 10. FANS performance under varying sample sizes on 50-node ER graphs. Performance degrades gracefully as the training set shrinks.

Samples N	F1 (Detection)	F1 (Dissection)
50,000	0.975	0.947
30,000	0.971	0.941
10,000	0.973	0.938
5,000	0.970	0.916

FANS maintains strong performance even when the training set is reduced by 90%. Detection F1 drops only modestly from 0.975 to 0.970, and dissection F1 remains above 0.91 throughout. This indicates that our framework is viable in practical settings where large observational datasets may not be available.

D.2.2. TRAINING EPOCHS

Table 11. FANS performance under varying training epochs on 50-node ER graphs. Performance remains strong even at 100 epochs.

Epochs	F1 (Detection)	F1 (Dissection)
100	0.966	0.937
500	0.977	0.940
3000	0.975	0.947

We next examine sensitivity to training duration by reducing the number of epochs from our default of 3000. Even with 100 epochs, detection F1 remains above 0.96 and dissection F1 above 0.93. This suggests that FANS does not require extensive training to achieve reasonable performance, which is useful when computational budgets are constrained.

D.2.3. MODEL CAPACITY

To assess robustness to model misspecification via underparameterization, we reduce the CNF’s hidden layer dimensions below the default of [32, 32, 32].

In contrast to the relative insensitivity to batch size, learning rate, and training duration, model capacity plays a more significant role. Reducing the hidden dimensions to [8, 8] leads to a noticeable drop in both detection and dissection performance, indicating that sufficient network expressiveness is required to capture the non-linear, non-additive structural equations considered in our framework. This aligns with the expected behavior of flow-based models: the CNF must be expressive enough to invert the underlying data-generating process, and insufficient capacity propagates estimation error into both the divergence statistic and the residual independence test.

Table 12. FANS performance under varying CNF hidden layer architectures on 50-node ER graphs. Unlike training hyperparameters, model capacity has a pronounced effect on performance.

Hidden Layers	F1 (Detection)	F1 (Dissection)
[32, 32, 32]	0.975	0.947
[16, 16]	0.917	0.864
[8, 8]	0.871	0.794

D.3. Robustness to Assumption Violations

D.3.1. VIOLATION OF ASSUMPTION

Assumption 3.3 (Generalized Affine Interaction) underpins the identifiability result between function-only and simultaneous shifts (Theorem 3.4). To evaluate robustness when this assumption is violated, we introduce a quadratic noise term controlled by a severity parameter λ , modifying the data-generating process as:

$$X_j = \sum_k \sin([PA_j]_k^2) + \sigma\left(\sum_k [PA_j]_k\right) \cdot \varepsilon_j + \lambda\left(\sum_k [PA_j]_k^2\right) \cdot \varepsilon_j^2, \quad (56)$$

where $\lambda = 0$ recovers our original setting and larger values induce stronger violations of the generalized affine interaction structure. As expected, performance degrades under assumption violations, since the standardized-noise argument in Theorem 3.4 no longer holds exactly.

Table 13. FANS performance for distinguishing function-only from simultaneous shifts under violations of Assumption 3.3. The severity parameter λ controls the magnitude of the violation.

Severity λ	F1 Score
0.0 (no violation)	0.622
0.5 (mild)	0.556
1.0 (severe)	0.467

D.3.2. ADDITIVE NOISE MODELS

Although FANS is designed for a broader class of non-additive structural equations, it should remain valid under the more restricted Additive Noise Model (ANM) setting. We evaluate this by setting the scale term $\sigma(\cdot) = 1$ in the data-generating process of Section 4.1 on 50-nodes ER graphs, reducing the SCM to an ANM.

Table 14. FANS performance on 50-node ER graphs under the ANM setting ($\sigma(\cdot) = 1$). Standard deviations across 30 DAGs are reported in parentheses.

	F1 (Detection)	F1 (Dissection)
FANS (ANM setting)	0.909 (0.084)	0.822 (0.206)

FANS achieves a moderate detection performance under the ANM setting. We attribute this to the fact that removing the parent-dependent scale term weakens the distributional signal distinguishing noise shifts from function shifts: when noise enters purely additively, a change in noise distribution produces a more subtle effect on the conditional distribution, making dissection inherently harder. This suggests that FANS benefits from heteroskedastic noise structure to perform dissection reliably.

D.3.3. ENVIRONMENT SWAP

FANS assumes that the causal DAG of the base environment (Environment 1) is known. To examine whether the choice of base environment affects performance, we swap the roles of Environments 1 and 2 on 15 datasets with 50-node ER graphs where the DAG structure is preserved across environments. In these cases, FANS achieves an F1 score of 0.953 for detection

and 0.979 for dissection. This indicates that, when the structural graph is shared, the assignment of base environment is not a source of bias in our procedure.

D.3.4. UNOBSERVED CONFOUNDING

Our framework assumes causal sufficiency, the absence of unobserved common causes. We briefly discuss the behavior of FANS when this assumption is violated. Consider an unobserved confounder U affecting both X and Y ($U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$). If the distribution of U shifts across environments, FANS will capture this variation through the observed residuals: the shift in U typically manifests as an apparent simultaneous mechanism shift on Y (as U enters Y 's structural equation) together with a noise shift on X (as the marginal of X changes). While this behavior does not recover the true underlying cause, the pattern could serve as a diagnostic signal for the presence of shifting hidden confounders.